Truss Grid Analyzer

User Guide

How to build a rig, load it, hang it and read the results

Written for version 1.28.0
12 October 2026

By **G.E. Simmons Falk**. Truss Grid Analyzer is an expansion, for indeterminate grids, of *Truss Load Analyzer - EOT* by **Delbert L. Hall and Jon Sogoian**, the originators of the program.

This guide is written for version 1.28.0 and is not updated for every release. Later versions may look or behave differently; the version you are running is shown next to the app name in the header. A searchable copy of this guide is supplied as an HTML file with the same name.

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1 About This Guide

Written for Truss Grid Analyzer version 1.28.0
12 October 2026

This guide explains how to use Truss Grid Analyzer: how to find your way around the screen, build a rig, put loads on it, hang it from hoists, and read and print the results. It is written for riggers, production managers and anyone else who plans truss rigs. You do not need to know structural engineering to follow it.

1.1 Which Version This Guide Describes

Everything here - the screenshots, button names, menu items and numbers - is taken from version **1.28.0**. The guide is **not** updated for every new release, so a later version may add features or look a little different. To see which version you are running, look next to the app's name at the top left of the screen, or open **File > About, method & version history**, which also lists what changed in each version since this one.

1.2 How to Use This Guide

- **New to the app?** Read the *Quick Start* (a separate short PDF), or chapters 2-6 of this guide, then work through Building a Box, Step by Step.
- **Looking for one thing?** In the HTML copy of this guide, type in the **Search** box at the top (or press `/`). In the PDF, use your PDF reader's search (`Ctrl` + `F`) or the bookmarks.
- **Something looks wrong?** See Warnings and Messages and Troubleshooting and Questions.

1.3 Conventions

Looks like	Means
`+ Truss`	A button in the app.
File > Save	Open the **File** menu and choose **Save**.
`Ctrl` + `Z`	Hold the first key and press the second. On a Mac, use `Cmd` where this guide says `Ctrl`.
Truss pieces	The name of a field, column or section on screen.
`12-6`	Something you type.

Screenshots use the **light** theme so they print clearly. The app starts in the **dark** theme; both show exactly the same things (see Dark or Light). Screenshots use feet and pounds; everything works the same in meters and kilograms.

> **Tip - boxes like this**
> Green boxes are tips, amber boxes are things to watch out for, and red boxes are safety points.

2 Introducing Truss Grid Analyzer

Truss Grid Analyzer works out the load on every chain hoist in a truss rig and checks every truss against its manufacturer's published limits. You describe the rig in four steps:

1. **Structure** - the trusses: their make and model, length, where they sit on the plan, and how they are bolted together at corner blocks.
2. **Loads** - what hangs on them: lights from a library of fixtures, or any weight you type, plus spread loads such as a drape or an LED wall.
3. **Hoists** - where the rig hangs from: chain hoists (or wire-rope dead hangs), their models and chain lengths.
4. **Results** - the load on every hoist, every truss check, and every warning.

Everything is recalculated the moment you change anything, so you can try ideas - move a hoist, add a light - and see the effect straight away.

2.1 What It Checks

- **Hoists:** the high hook load (the truss load on the hook plus the hoist, its chain and hardware) against the hoist's rated capacity, and the dynamic load (static × a factor for the hoist's speed).
- **Truss spans:** the point loads on every span against the manufacturer's load table for that span length, with the ANSI repetitive-use factor (0.85) unless the table already includes it.
- **Cantilevers:** the overhang past the last support: no longer than a quarter of the maximum span, and no more load than the table allows.
- **Bending moment and shear:** the whole length of each truss, which catches bending over supports and heavy loads next to a support that a span-by-span check misses.
- **Deflection:** how far each span sags, against the maker's published limit where there is one (otherwise a default of span/160).
- **The rig as a whole:** a *whole-rig analysis* works out how trusses bolted into a grid share load, including hoists that go slack and assemblies that would tip.
- **Input mistakes:** a load with no weight, two hoists at one point, bolted ends that do not meet, parts from different truss families bolted together, and more.

2.2 What You Need to Know First

> **A calculation aid, not an engineering approval**
> This tool is intended for entertainment rigging professionals to help them make rigging decisions. Every attempt has been made to be accurate, but the authors are not responsible for errors in the program or the results of its use. **You are responsible for verifying the results before using them to make rigging decisions.** Check results against the manufacturer's current data, and have a qualified person review every rig before it flies.

- The results are only as good as what you enter. Weigh or look up every load, and include cable, clamps and safeties (the app has allowances for these - see Rig Settings).
- Some figures are **estimates** and the app says so: truss stiffness, the allowable bending moment and shear, and deflection are worked out from the manufacturers' load tables, not published values.
- The app does not check the structure the hoists hang from (beams, roof steel, a ground-support system) - only the hoists and the trusses.

> **Indeterminate rigs (ANSI E1.6-3)**
> ANSI E1.6-3 (*Selection and Use of Serially Manufactured Chain Hoists in the Entertainment Industry*, section 2.11) calls a load system **indeterminate** when the way its load divides between the hoists depends on the stiffness of the structure as well as on where the loads and hoists are. In practice that is any rig on more than two hoists in a straight line, or more than three hoists not in a line: a truss on three hoists, and almost every grid or box. On these rigs load can shift between hoists as they run or if one is out of level, and the standard warns that stiff indeterminate structures can shift load very quickly.
>
> Section 7.1 of the standard accepts a load calculation as a way to reduce the risk of overload only when all the weights are known and accounted for, an accurate calculation method is used, and any load shifting will not be enough to cause an overload. The whole-rig analysis in this app is built for these rigs - it solves the grid with hinged, semi-rigid and rigid joints and uses the worst - but it is still an estimate. Allow a margin for load shifting, level hoists carefully, and verify the loads (for example with load cells) where the margin is small.

2.3 Where It Came From

Truss Grid Analyzer, by G.E. Simmons Falk, is an expansion of *Truss Load Analyzer - EOT*, the Excel workbook created by **Delbert L. Hall and Jon Sogoian**, who are the originators of the program, for **indeterminate grids**. The workbook works truss by truss; Truss Grid Analyzer extends it to trusses bolted into grids and boxes and to trusses on three or more hoists - rigs whose hoist loads depend on the stiffness of the trusses and joints, not on the load positions alone (see *Indeterminate rigs* above). It uses their method and their truss, hoist and fixture data, adds manufacturers' own load tables, and adds the grid model, the whole-rig analysis, and the plan and 3D views. It is free, and must never be sold (see Credits, License and Disclaimer).

3 Getting Started

3.1 What You Need

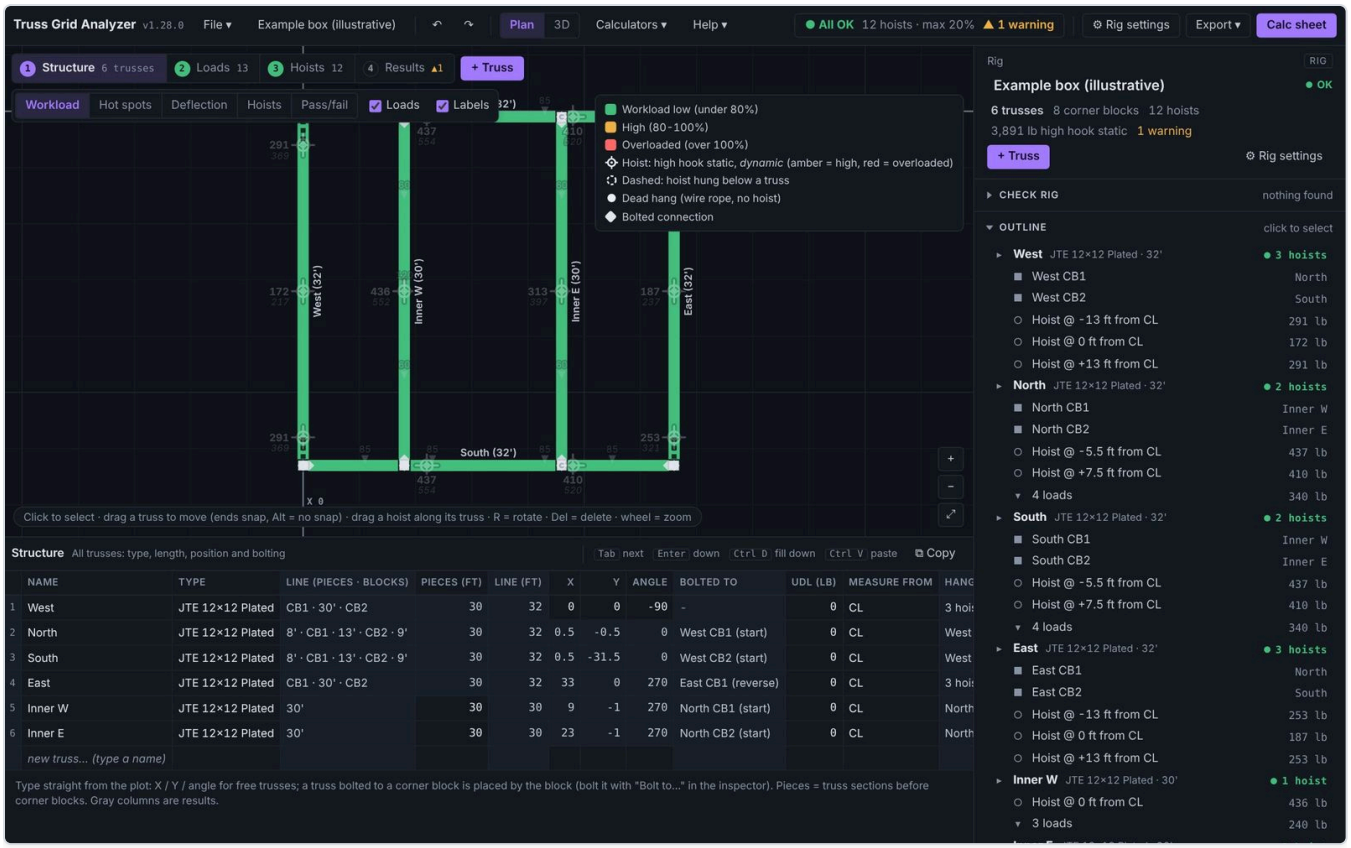
- A computer with a current web browser: Google Chrome, Microsoft Edge, Mozilla Firefox or Apple Safari. A large screen (a laptop or bigger) works best; the app is not designed for phones.
- The app file. The whole app is a single file called `index.html` - on the project's GitHub page it is `dist/index.html`. It contains everything: the program, the truss, hoist, corner block and fixture databases.

There is nothing to install, no account, and no internet connection is needed. The app never sends your rig anywhere. (The only things that go online are the links to manufacturers' data sheets, which open in a new browser tab when you click them.)

3.2 Opening the App

1. Save `index.html` somewhere you will find it again, for example a *Truss Grid Analyzer* folder in Documents. You can rename it (for example `Truss Grid Analyzer 1.28.0.html`).
2. Double-click it. It opens in your default browser. (Or drag it onto an open browser window.)
3. Check the version at the top left, next to **Truss Grid Analyzer**.

The first time you open it, the app shows the **example box** - a small box truss with two inner trusses, loads and twelve hoists - so you have something to explore. You can bring it back any time with **File > Load example box**.



The app as it first opens: the example box, in the dark theme.

3.3 Where Your Work Is Kept

The app saves **every change automatically** inside the web browser on your computer. Close the browser, come back tomorrow, and your rig (and your custom trusses, hoists and fixtures) will be there.

> **Automatic saving is not a backup**
>
> - It is kept by **this browser on this computer**. Another browser, another computer or a private/incognito window starts empty (with the example box).
> - Clearing the browser's history or "site data" can erase it.
> - It holds **one rig at a time** - the one on screen. Starting a new rig or opening another replaces it (Undo brings it back only until you close or reload the page).
>
> So keep every rig you care about as a file with **File > Save** (see Files, Export and Sharing).

3.4 Starting a New Rig

Choose **File > New rig** and click **OK**. The plan is cleared and you start at step 1.

> **Save first**
> **File > New rig**, **File > Open...** and **File > Load example box** each replace the rig on screen. Each is one Undo step, so `Ctrl`+`Z` brings the old rig back - but only until you close or reload the page. Use **File > Save** before any of them to keep a copy.

3.5 Naming the Rig

Click the rig's name in the top bar (a new rig is called *Untitled rig*), type a new name and click **OK**. You can also edit it at the top of the rig panel (click an empty part of the plan to show it). The name is used for the saved file, the calculation sheet and exported pictures.

3.6 Dark or Light

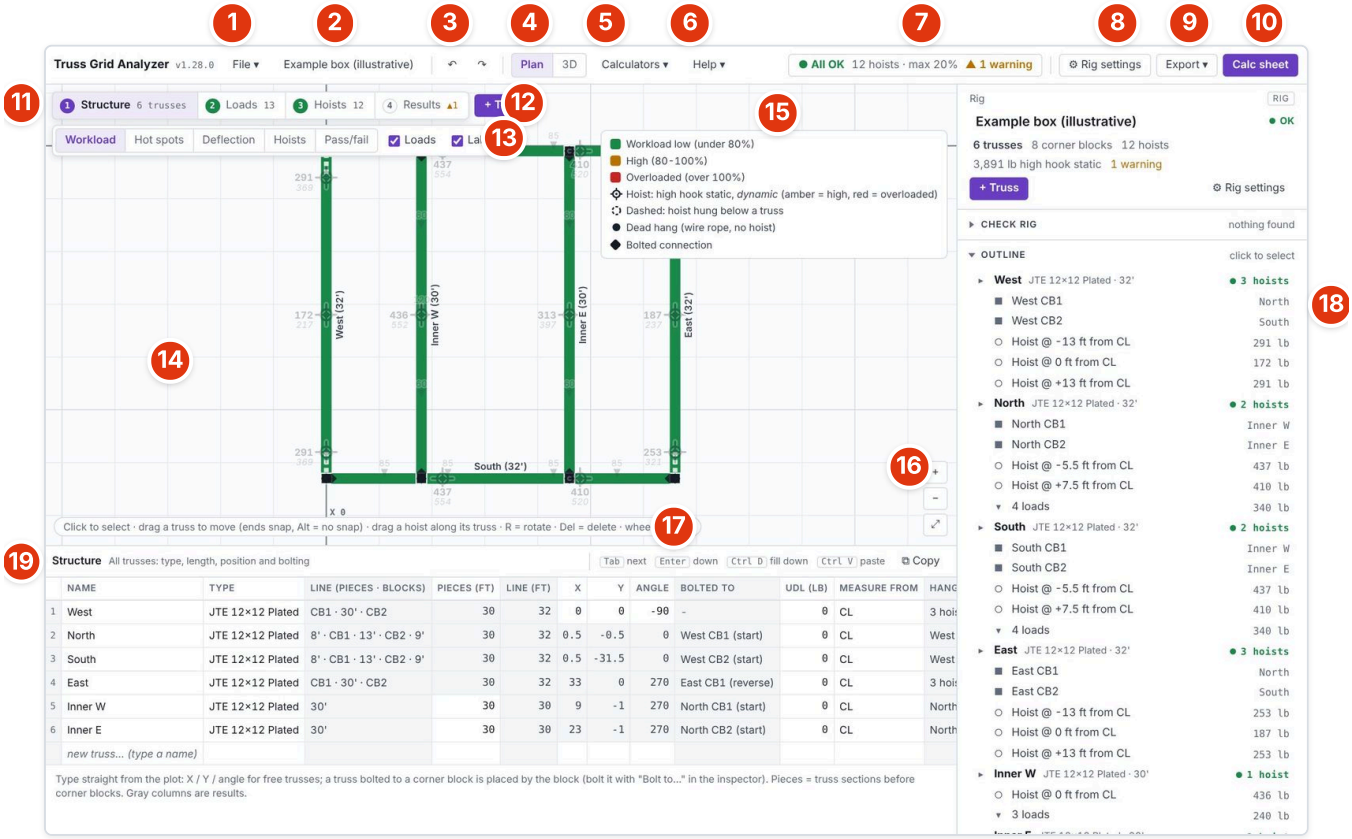
The app starts in a dark theme, which is easy on the eyes in a dark venue. Choose **File > Theme > Light** for daylight or an office, or **Follow system** to match your computer's setting. The same choice is in Rig Settings. The theme is remembered on this computer; it is not saved in rig files and does not change any result.

3.7 Undo and Redo

Almost every change can be undone: click ↶ / ↷ in the top bar, or press `Ctrl`+`Z` to undo and `Ctrl`+`Y` (or `Ctrl`+`Shift`+`Z`) to redo. The app keeps the last 100 changes. A paste, a multi-hoist edit, New rig, Open and Load example box each count as one change. The history is lost when you close or reload the page. (Undo does not work while you are typing in a box - finish the entry first.)

4 The Screen

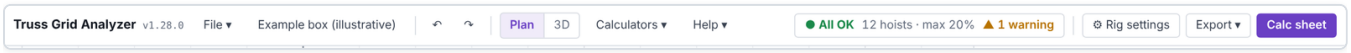
The app has one main screen. The picture below shows it with the example box loaded and nothing selected.



The main screen.

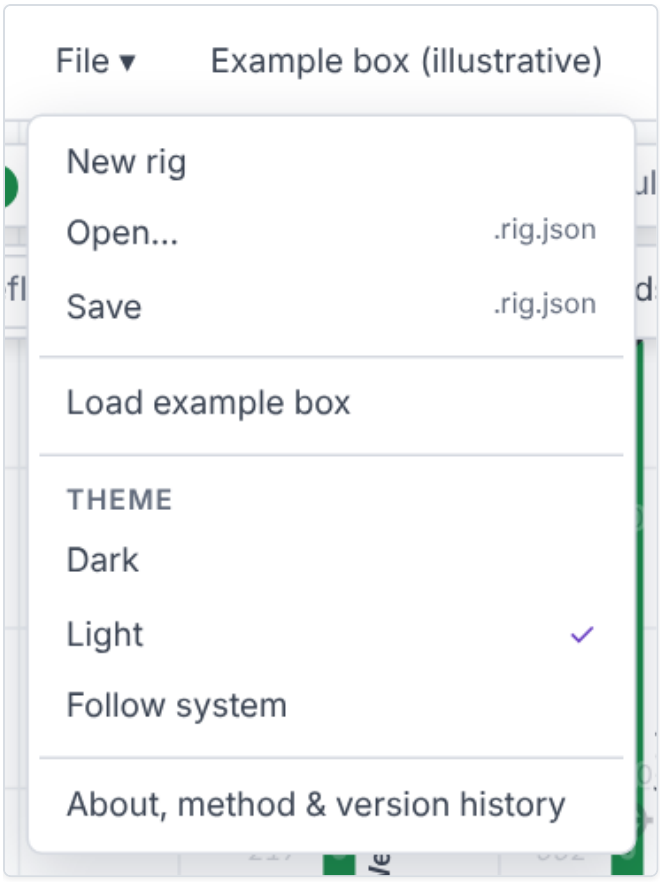
1. **File menu** - new, open, save, example box, theme, About.
2. **Rig name** - click to rename.
3. **Undo / redo.**
4. **Plan / 3D** - which view is shown.
5. **Calculators menu** - standalone tools and the databases.
6. **Help menu** - the Quick Start and this User Guide.
7. **Rig status** - click it for the results.
8. **Rig settings** - rig-wide options.
9. **Export menu** - hoist table, picture, model, print.
10. **Calc sheet** - the printable calculation.
11. **Steps bar** - 1 Structure, 2 Loads, 3 Hoists, 4 Results.
12. **Add button** - changes with the step: + Truss, + Load, + Hoist, Calc sheet.
13. **View toolbar** - color mode, show loads, show labels.
14. **Plan** - the rig to scale; click, drag, zoom.
15. **Key (legend)** - what the colors and symbols mean.
16. **Zoom** - zoom in, zoom out, fit.
17. **Hint bar** - what the mouse and keys do on the plan.
18. **Side panel** - details of whatever is selected.
19. **Grid** - a spreadsheet for the current step.

4.1 The Top Bar

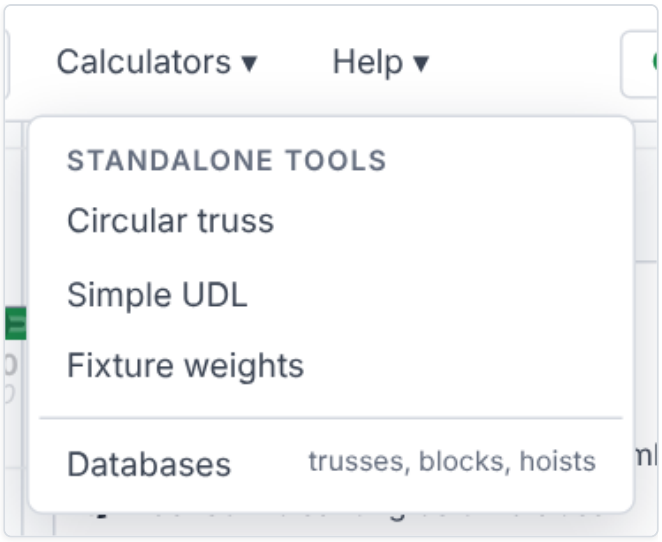


The top bar.

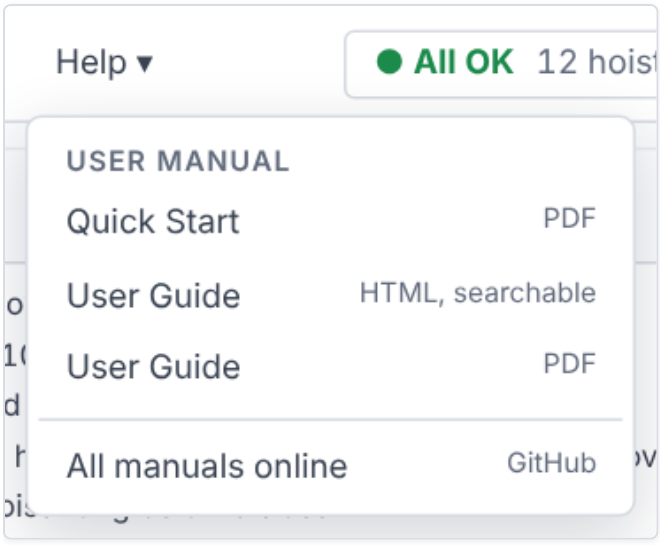
Item	What it does
Truss Grid Analyzer v1.28.0	The app's name and version.
File ▾	**New rig**, **Open...** a `.rig.json` file, **Save** the rig as a file, **Load example box**, **Theme** (Dark / Light / Follow system), and **About, method & version history**.
Rig name	Click to rename the rig.
↶ ↷	Undo and redo.
Plan | 3D	Show the rig as a plan (from above, to scale, editable) or in 3D (for looking).
Calculators ▾	**Circular truss**, **Simple UDL**, **Fixture weights** and **Databases** (trusses, blocks, hoists). See Calculators and Databases.
Help ▾	**Quick Start** (PDF) and the **User Guide** (searchable HTML or PDF), opened in a new tab from the `docs/manual` folder next to the app, and **All manuals online** (the same folder on GitHub - use it if you have only the app file). See Help and Search in This Guide.
Status	A one-line summary: **● All OK** or **● 2 to check**, the number of hoists and the highest hoist workload, and **▲ n warnings**. Click it to go to step 4 (the Warnings tab if there are warnings).
⚙ Rig settings	Opens the rig settings page (units, dynamic factor, allowances, deflection limit, paper...). See Rig Settings.
Export ▾	**Copy hoist table** (for Excel), **Save view as PNG**, **Export whole-rig model**, **Print screen**.
Calc sheet	Opens the printable calculation sheet. See The Calculation Sheet.



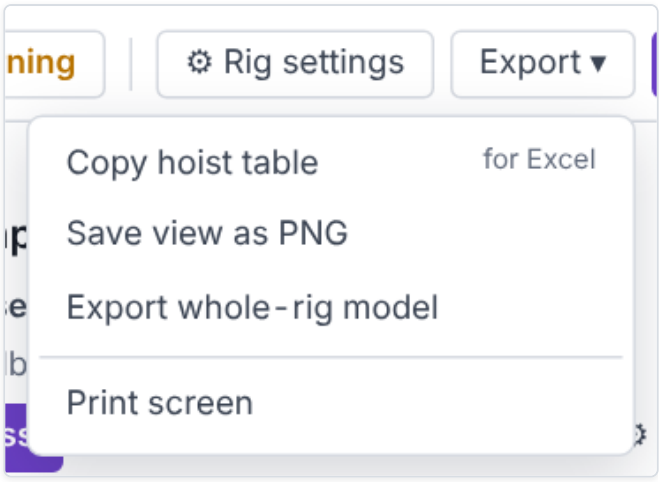
The File menu.



The Calculators menu.

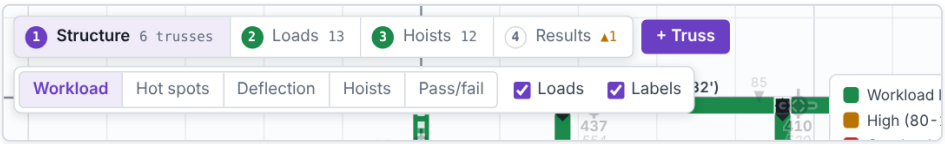


The Help menu.



The Export menu.

4.2 The Steps Bar



The steps bar, the add button and the view toolbar.

The four steps split the work into manageable parts. Click a step (or press `1`, `2`, `3` or `4`) to change the grid under the plan and the add button. You can go to any step at any time and in any order - the steps do not lock anything. Each step shows a count: the number of trusses, loads and hoists, and on **Results** a tick (✓, all OK), **▲n** (n warnings) or **✕n** (n things overloaded or unsolved). The step number turns green when that step has something in it (for Hoists: when there are hoists and nothing is left to check); the step you are on is highlighted.

The **add button** next to the steps does the obvious thing for the step: **+ Truss** (step 1), **+ Load** or **+ Hoist** (steps 2 and 3, on the selected truss - if no truss is selected it adds a new truss instead), and **Calc sheet** (step 4).

4.3 The Plan

The plan shows the rig from above, drawn to scale on a grid, with north at the top. The thin dark lines marked **X 0** and **Y 0** are the origin. Each truss is drawn as a colored bar with its name and length; corner blocks are dark squares; hoists are rings with cross-hairs, labeled with their loads; loads are small triangles labeled with their weight. See Plan and 3D Views for everything you can do on it.

4.4 The Side Panel

The panel on the right shows whatever you last clicked:

You clicked	The panel shows
An empty part of the plan	The **rig**: name, status, totals, Check rig, an outline of every truss, corner block, hoist and load, and a summary of the rig settings.
A truss	The **truss**: type, length, corner blocks, placement and bolting, loads, hoists, checks, and advanced options.
A corner block	The **corner block**: its faces and what is bolted to each, its type and weight, what it carries.
A hoist	The **hoist**: its loads, where the load comes from, model, chain, dynamic factor, position.
A load (in a list or grid)	The **load**: item, weight, clamp, category, note, position, mirror.

At the top of every panel is a trail such as **Rig › North › Hoist @ -5.5 ft from CL**: click **Rig** or the truss name to go back up. Each panel is split into sections you can open and close by clicking their titles; **Collapse all** and **Expand all** do all of them at once, and the app remembers which ones you like open.

4.5 The Grid

Under the plan is a spreadsheet for the current step: every truss (step 1), every load grouped by truss (step 2), or every hoist grouped by truss (step 3), with results updating as you type. Step 4 shows the results tables instead. The grid works like Excel - see Working in the Grids.

4.6 Resizing the Panels

Drag the border between the plan and the side panel, or between the plan and the grid, to make either bigger. Double-click a border to put it back. The app remembers your layout on this computer.

5 Key Ideas

A few words have a specific meaning in the app. The Glossary has the full list.

5.1 Trusses and Truss Lines

In the app, a **truss** is one straight line of truss - which may be several sections (pieces) bolted end to end, and may include corner blocks along it. For example a 32 ft line might be a corner block, 30 ft of truss in 10 ft and 5 ft sticks, and another corner block. The app treats the whole line as one continuous beam.

Each truss has a **start** and an **end**. On the plan, the start is at the truss's X, Y position, and the truss runs from there in the direction of its angle: 0° runs east (to the right), 90° north (up), 180° west and 270° south.

5.2 Corner Blocks and Bolting

A **corner block** is a short block that trusses bolt to at right angles (a 6-way block has six faces: four sides, top and bottom). In the app a corner block always **belongs to one truss** - it is part of that truss line, at its start, its end or somewhere along it. Other trusses are then **bolted to** the block. A bolted truss is positioned by its block: it moves when the block's truss moves, and you cannot drag it on its own.

A truss can be **free** (you place it by typing its X, Y and angle, or dragging it) or **bolted** (placed by a corner block). A truss can also be **stacked** on top of another truss or **clamped** below it at the point where they cross.

5.3 Measuring Positions: Start, Centerline or End

Every load and hoist position is a distance along its truss, measured from one of three places:

- **start** - from the start of the whole line (0 = the start end);
- **centerline** (CL) - from the middle, negative toward the start and positive toward the end (`-12` , `+12`);
- **end** - from the far end of the line.

Each truss has a **Measure from** setting (a new truss uses the centerline), and each load or hoist can use its own. Pick whichever matches the drawing you are working from. Lengths are measured along the whole line, including corner blocks.

5.4 Loads

A **load** is a point load at one position: a fixture from the library (with its clamp), or any weight you type. A **mirrored** load also puts the same weight at the mirror-image position on the other side of the centerline, and follows any change you make to the original. A **UDL** (uniformly distributed load) is a weight spread evenly over the whole length of a truss - a drape, an LED wall hung along it, or cable.

5.5 Hoists, Low Hook and High Hook

A **hoist** is a support point where a chain hoist holds the truss (or, as a **dead hang**, a wire rope with no hoist). The app uses these words throughout:

Term	Meaning
Low hook load	What the truss hangs on the hoist's hook (sometimes written R, the reaction).
High hook load	The low hook load + any **Add %** + the weight of the hoist, its chain and hardware: what the structure above carries.
Static / dynamic	Static is the load hanging still. Dynamic is static × the **dynamic factor** for the hoist's movement (for a hoist with a listed speed: speed in ft/min ÷ 60 + 1, so a 16 fpm hoist is 1.267).
Workload	A load as a percentage of what is allowed: for a hoist, high hook static ÷ rated capacity; for a truss, the highest of its span, cantilever, moment, shear and deflection checks.
Level sensitivity	How much a hoist's load changes if it runs 1/4" (6 mm) high or low compared with the others.

5.6 Spans and Cantilevers

A **span** is the part of a truss between two supports (hoists or bolted connections). A **cantilever** is the part past the last support at either end. Manufacturers publish how much load each span length can carry; the app reads those tables. A cantilever may be no longer than a quarter of the truss's maximum span.

5.7 The Whole-Rig Analysis

When trusses are bolted into a grid, they share load: a truss sags a little, and some of its load moves to neighboring hoists. The app works this out with a **whole-rig analysis** (a stiffness or finite-element analysis of the grid). Because nobody publishes how stiff the joint through a corner block is, the app solves the rig several ways - with the joints **hinged**, **semi-rigid** and **rigid** - and every check uses the worst result. You do not need to do anything to make this happen; see How the App Calculates if you want to know more.

5.8 Statuses

Status	Used for	Meaning
OK	trusses, hoists	Within every limit. "OK - high" on a truss means 80% or more.
Overloaded	trusses, hoists	Over a limit.
Too long	trusses	A span or cantilever longer than the table allows.
Too long and overloaded	trusses	Both.
Slack	hoists	The load would push the hoist up; its chain goes slack and it carries nothing.
Unstable	trusses, hoists	Not held up properly - it would tip. Its numbers must not be used.
Check	trusses (panel)	A check has failed or a warning applies; open its Checks section.
Not solved	trusses	It has no supports yet, or depends on something missing or on a loop.

6 Entering Values

6.1 Lengths

Any box that takes a length accepts several ways of writing it. In imperial units:

You type	The app reads
`12.5`	12.5 ft (decimal feet)
`12-6`	12 ft 6 in
`12'6"` or `12' 6"`	12 ft 6 in
`12' 6 1/2"` or `12-6-1/2`	12 ft 6½ in
`12 ft 6 in`, `12 ft`	12 ft 6 in, 12 ft
`6"` or `6 in`	6 in (0.5 ft)
`-3-6`	minus 3 ft 6 in (a position before the centerline)

In metric units, type meters (`3.8`), or add a unit: `380 cm` , `3800 mm` . Feet-and-inches still work.

If the app cannot read what you typed, the box flashes red and keeps its old value. How lengths are *shown* (decimal feet such as 4.1667, or feet-inches such as 4'-2") is set in Rig Settings > **Show lengths as**.

6.2 Weights and Other Numbers

Weights are typed as plain numbers in the rig's units (pounds, or kilograms in metric). In the load grid's **Item** cell and the quick-add box you can include the unit: `45 lb` or `20 kg` . Other units the app uses: lb/ft (kg/m) for weight per length, inches (mm) for level offsets and truss widths, lb/in (kg/mm) for hoist stiffness, ft/min (m/min) for hoist speed, lb-ft (kg·m) for moments.

6.3 Metric or Imperial

Choose **Units** in Rig Settings. Switching changes only what is shown and typed: the rig is stored and calculated the same way, so no result and no truss or hoist choice changes. The setting is saved with the rig. Trusses whose manufacturer data is metric are always checked against their own metric table.

6.4 Committing a Value

A value takes effect when you press `Enter` or `Tab` , or click somewhere else. Results update straight away.

7 Working in the Grids

Steps 1, 2 and 3 each have a spreadsheet-style grid under the plan. You can build a whole rig from the grids alone - typing straight from a drawing or pasting from Excel - or use them alongside the plan and side panel. Every edit in a grid is exactly the same as making it in the side panel, and each edit (or paste) is one Undo step.

7.1 Moving Around and Typing

To	Do this
Select a cell	Click it. The cell gets a blue outline and its row's truss, load or hoist is selected on the plan and in the side panel.
Move	Arrow keys; `Tab` / `Shift`+`Tab` (right / left, skipping gray result columns); `Enter` / `Shift`+`Enter` (down / up). After tabbing along a row, `Enter` goes back to the column you started in, on the next row - like Excel.
Edit	Just start typing (replaces the value), or double-click, press `F2`, or click an already-selected cell (edits the value).
Finish an edit	`Enter` (and move down), `Tab` (and move right), or click elsewhere. `Esc` cancels.
Tick a checkbox	Click it, or press `Space` on it.
Copy the cell above	`Ctrl`+`D` (fill down).
Copy a cell	`Ctrl`+`C`. The **⧉ Copy** button in the grid's header copies the whole grid as tab-separated text for Excel.
Paste	`Ctrl`+`V` - see below.
Delete a row	`Delete` (or `Shift`+`Backspace`). Deleting a truss asks first.
Stop editing the grid	`Esc` (with no edit open) - the plan's keys work again.

Gray columns are results or values set elsewhere (for example the position of a bolted truss); you cannot type in them. A column that is gray for one row may be editable in another.

7.2 Adding Rows

The last row of each group is a "new" row in italics: *new truss... (type a name)*, *+ load on North (start typing)*, *+ hoist on West (type its position)*. Click one of its editable cells and type: a new truss, load or hoist is created with what you typed, and the cursor stays on the new row so you can keep going.

7.3 Picking from a List

Some cells search a list as you type - the truss **Type**, the load **Item / fixture** and the **Hoist** model. Type any part of the name (several words narrow it down: `jte 12 plated`, `mac aura`, `lode 1t`). Use the arrow keys to choose, then `Enter` to pick, or `Tab` to pick and move to the next cell. `Esc` closes the list.

7.4 Pasting from a Spreadsheet

Copy a block of cells in Excel (or any spreadsheet), click the cell where the top-left value should go, and press `Ctrl` + `V`. Each line fills a row, from that cell to the right and down. If you run past the last row of a group, new rows are added to it. Empty pasted cells leave the value alone; cells the app cannot read are skipped and it tells you how many. Paste one value (no tabs) to put it in the selected cell.

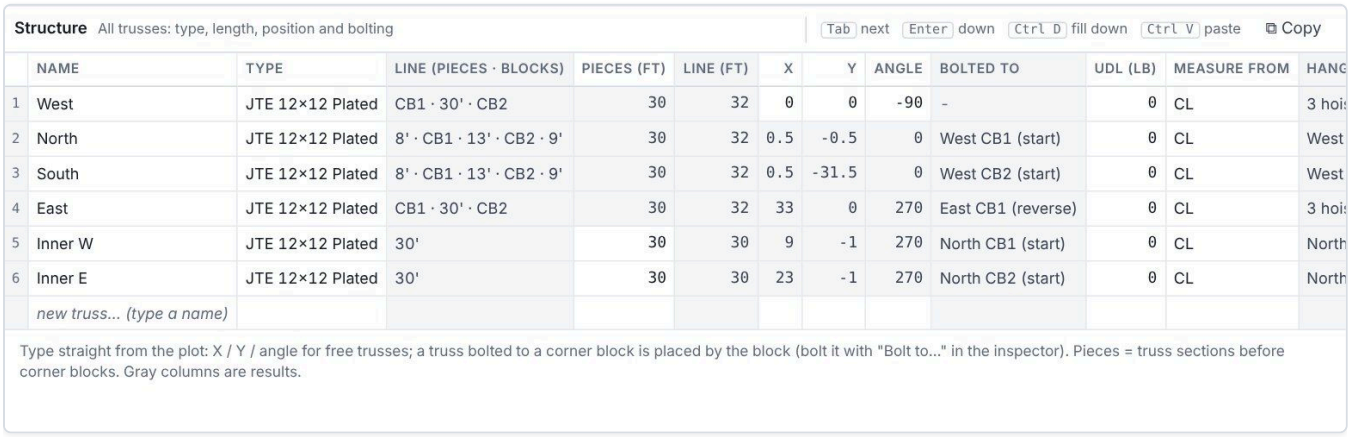
> **Column order for pasting**
> **Loads:** At, From, Item, Weight, Clamp, Mirror, Note, Category (Mirror: `yes`, `y`, `x` or `1`).
> **Hoists:** At, From, Hoist, Chain / rope, DLF, Hardware, Hangs from, Level.
> Start the paste in the **At** column. Put the column you start in first in your spreadsheet selection.

7.5 Choosing Several Hoists at Once

The Hoists grid (step 3) lets you change many hoists in one go - see Editing Many Hoists at Once.

8 Step 1 - Structure

In step 1 you create every truss, set its make, model and length, add its corner blocks, place it on the plan and bolt the trusses together. Build all the trusses first; loads and hoists come later.



Structure All trusses: type, length, position and bolting

Tab next Enter down Ctrl D fill down Ctrl V paste ⧉ Copy

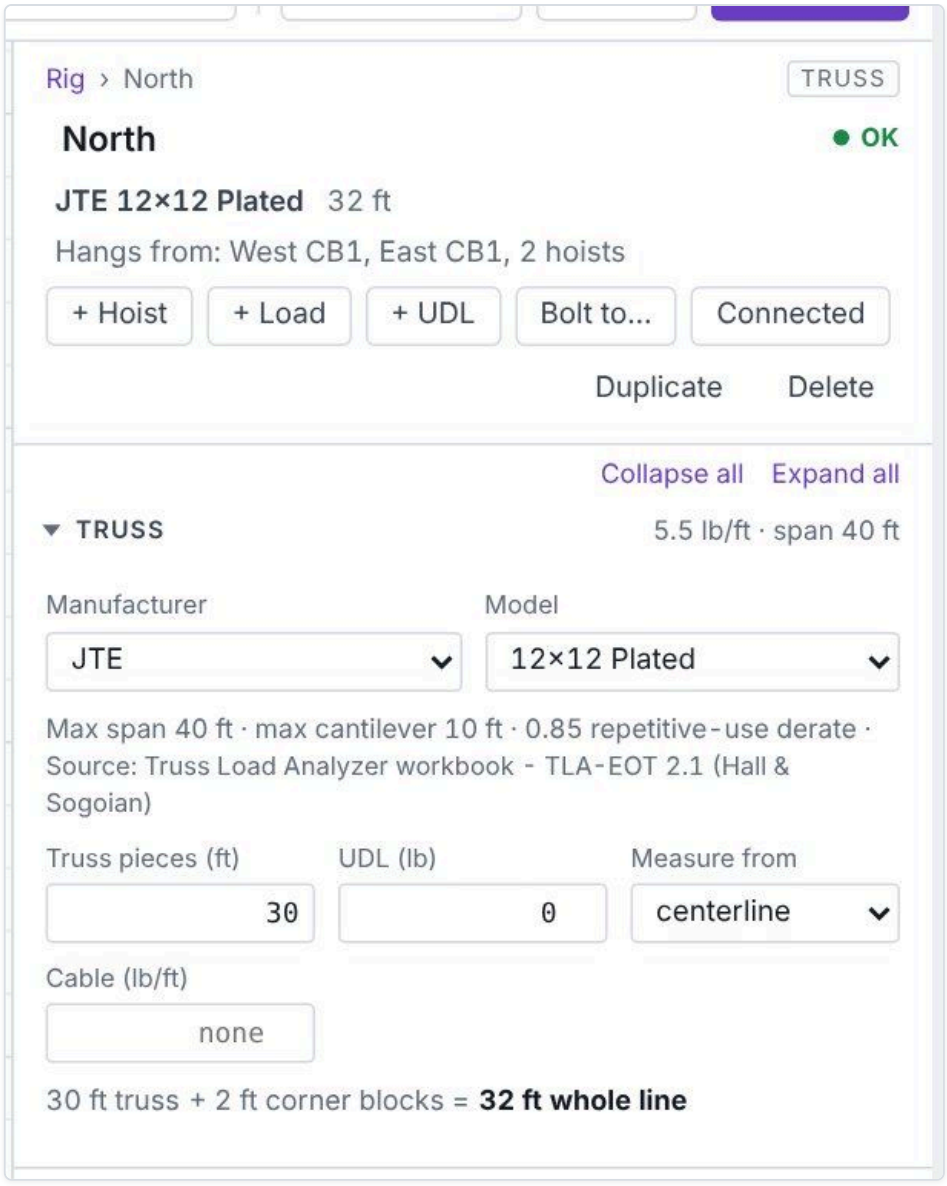
	NAME	TYPE	LINE (PIECES · BLOCKS)	PIECES (FT)	LINE (FT)	X	Y	ANGLE	BOLTED TO	UDL (LB)	MEASURE FROM	HANG
1	West	JTE 12×12 Plated	CB1 · 30' · CB2	30	32	0	0	-90	-	0	CL	3 hoi
2	North	JTE 12×12 Plated	8' · CB1 · 13' · CB2 · 9'	30	32	0.5	-0.5	0	West CB1 (start)	0	CL	West
3	South	JTE 12×12 Plated	8' · CB1 · 13' · CB2 · 9'	30	32	0.5	-31.5	0	West CB2 (start)	0	CL	West
4	East	JTE 12×12 Plated	CB1 · 30' · CB2	30	32	33	0	270	East CB1 (reverse)	0	CL	3 hoi
5	Inner W	JTE 12×12 Plated	30'	30	30	9	-1	270	North CB1 (start)	0	CL	North
6	Inner E	JTE 12×12 Plated	30'	30	30	23	-1	270	North CB2 (start)	0	CL	North
	new truss... (type a name)											

Type straight from the plot: X / Y / angle for free trusses; a truss bolted to a corner block is placed by the block (bolt it with "Bolt to..." in the inspector). Pieces = truss sections before corner blocks. Gray columns are results.

The Structure grid: one row per truss.

8.1 Adding a Truss

Click **+ Truss** (above the plan, or in the rig panel), or type a name in the grid's *new truss...* row. A new truss is a 20 ft JTE 12×12 Plated, named *Truss 1*, *Truss 2*..., measured from its centerline, with no loads and no hoists, placed at X 0, Y 0 running east. It is selected, so the side panel shows it.



The top of a truss's panel: name, status, type, length, what it hangs from, and actions.

The top of the truss panel shows:

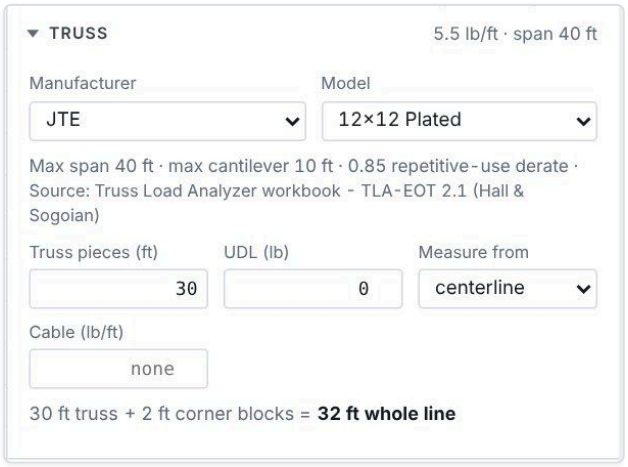
- the **name** (click it to rename) and its **status**;
- its type and length, and **Hangs from**: the trusses or corner blocks it is bolted to and how many hoists it has;
- buttons: `+ Hoist` (adds a hoist in the middle), `+ Load` (jumps to the quick-add box), `+ UDL` (the same, for a UDL), `Bolt to...`, `Connected`, rotate `⟳` (free trusses only), **Duplicate** and **Delete**.

Connected dims everything on the plan that is not joined to this truss (bolted, stacked, or hung on a hoist below it) - a quick way to spot a truss you meant to bolt on but did not. Click it again to turn it off.

8.2 Choosing the Truss Type

Open the **Truss** section. Choose the **Manufacturer**, then the **Model**.

- Models marked **(MFG)** use the manufacturer's own current load table and are listed first. The others come from the original Truss Load Analyzer workbook. Where a workbook row allows more than the maker's current table for a span in your rig, the app warns and names the maker's row to use instead.
- **(metric data)** means the maker publishes a metric table; it is checked in whole-meter steps.
- **(custom)** is a truss you added yourself (see Databases).



The Truss section.

Under the pickers the app shows the maximum span, the maximum cantilever (a quarter of the maximum span), the repetitive-use factor applied, and where the data comes from. For manufacturer data, **maker's load table ↗** opens the maker's own document in a new tab (this needs an internet connection).

In the grid, click the **Type** cell and type part of the name (`tyler gt` , `christie a`) instead.

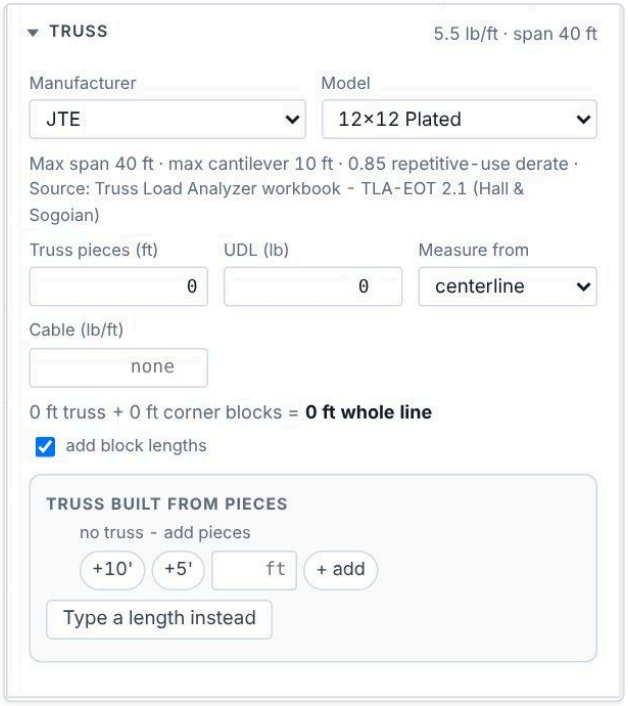
8.3 Setting the Length

A truss line is its **truss pieces** plus any **corner blocks** in the line:

- **Truss pieces** - the total length of truss sections, not counting corner blocks. Type it (`30` , `28-6`).
- The line under the fields adds it up: *30 ft truss + 2 ft corner blocks = 32 ft whole line*. With **add block lengths** ticked (the default) each corner block's length is added to the pieces. Untick it if the length you typed already includes the blocks.

Building a Truss from Standard Pieces

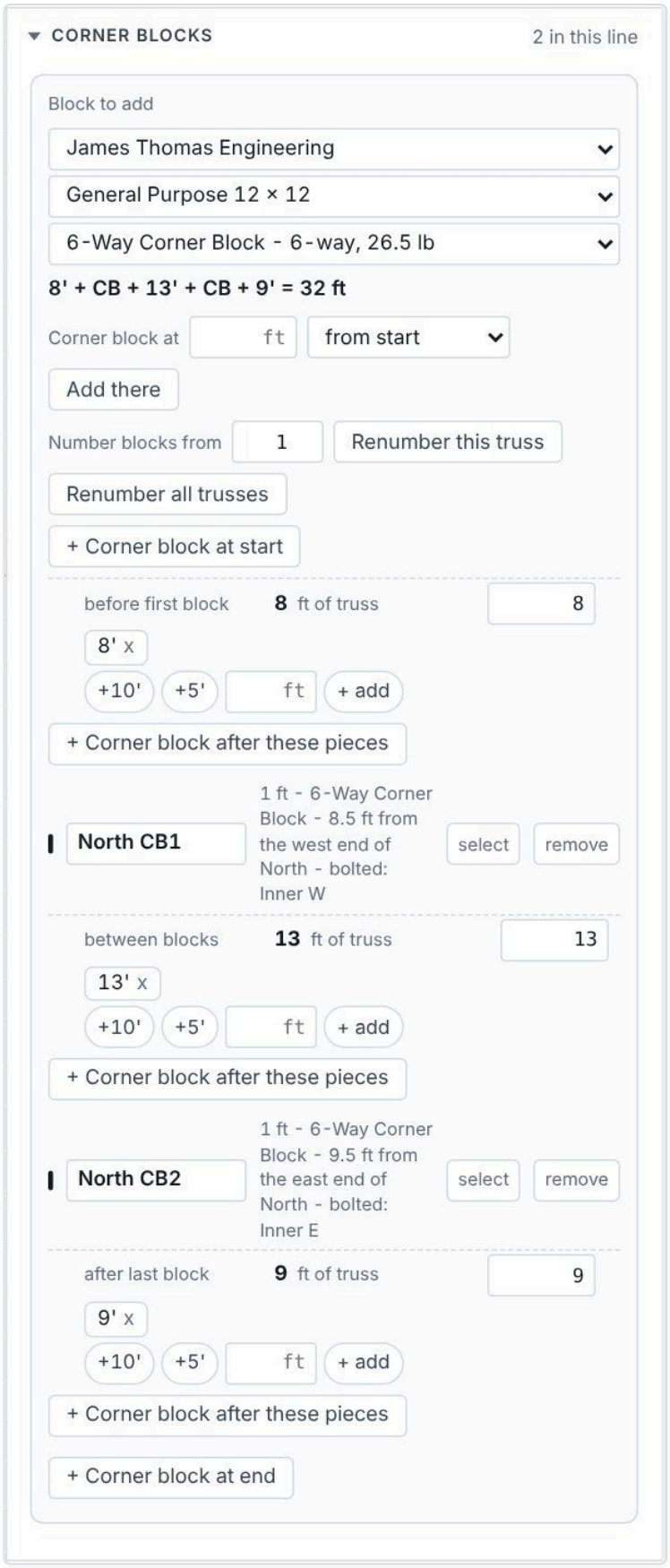
Click **Build from pieces...** to assemble the line from stock lengths. Buttons such as **+10'** and **+5'** add a stick of that length (the lengths offered depend on the truss family); type another length in the small box and click **+ add** for an odd piece (the app remembers it for next time). Each piece is shown as a tag - click a tag to remove that piece. **+CB** between two tags puts a corner block between those pieces. **Type a length instead** goes back to a single length.



A truss built from pieces.

8.4 Corner Blocks in a Truss Line

Open the **Corner blocks** section. First choose the **Block to add** - maker, truss family and block (with its number of ways and weight). The app suggests a block that fits the truss.



The Corner blocks section of a truss that has two blocks in its line (8' + CB + 13' + CB + 9').

Then add blocks in one of these ways:

Button	Puts a corner block...
+ Corner block at start / at end	at that end of the line (the line gets longer by the block, if **add block lengths** is ticked).
Corner block at ... from ... > Add there	with its center at an exact distance from the start, centerline or end; the truss is split there.
Set segment lengths...	(once there is a block in the line) switches to typing each run of truss between blocks: *before first block*, *between blocks*, *after last block*. Each run can be typed as a length or built from pieces.
+ Corner block after these pieces	after a run of truss (in segment mode).
+CB	between two pieces (in the piece tags).

The line is summed up as, for example, **8' + CB + 13' + CB + 9' = 32 ft**. Each block gets a row with its name (editable), its length, type and position, and what is bolted to it; **select** opens the block's own panel and **remove** takes it out of the line. Blocks are named after the truss (*North CB1*, *North CB2*...); **Number blocks from** with **Renumber this truss** or **Renumber all trusses** renames them in order along each line.

8.5 Placing a Free Truss

A truss that is not bolted to anything is **free**. Place it in any of these ways:

- Type **Plan X**, **Plan Y** (the start point) and **Angle** in the **Placement** section, or in the grid's X, Y and Angle columns. Angles: 0 = east, 90 = north, 180 = west, 270 (or -90) = south.
- Drag it on the plan. It moves in half-foot steps, and its ends snap to the ends of other trusses within 1.5 ft; hold `Alt` while dragging to stop the snapping.
- Press `R` (or click ⟳) to turn the selected truss 90° counterclockwise about its start.

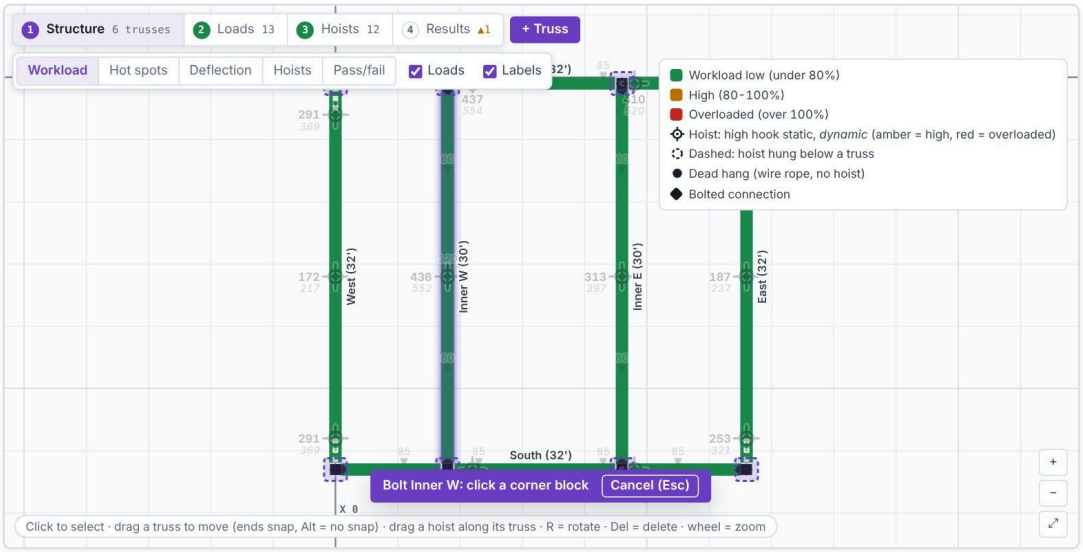
To move a whole rig so it is centered on the origin, use **Center on 0,0** in Rig Settings.

8.6 Bolting a Truss to a Corner Block

Bolting joins two trusses: the truss you bolt is placed by the block, at right angles to the block's truss, and follows it from then on.

1. Make sure the corner block exists: it must be in another truss's line (see above).
2. Select the truss you want to bolt on. In **Placement > Bolt to a corner block**, check the two options (the defaults are usually right):
   - **end: automatic**, **its start meets the block**, **its end meets the block**, or **the block is along it** (the block sits partway along this truss, as where a truss passes through a block);
   - **direction: automatic** or **extends north / south / east / west** - which way the truss runs from the block.

3. Click **Bolt to...** (in the Placement section or at the top of the panel). A blue bar appears: *Bolt North: click a corner block*, and the corner blocks you can use are outlined.
4. Click the corner block on the plan. The truss jumps into place. (Press `Esc` or **Cancel** to stop without bolting.)



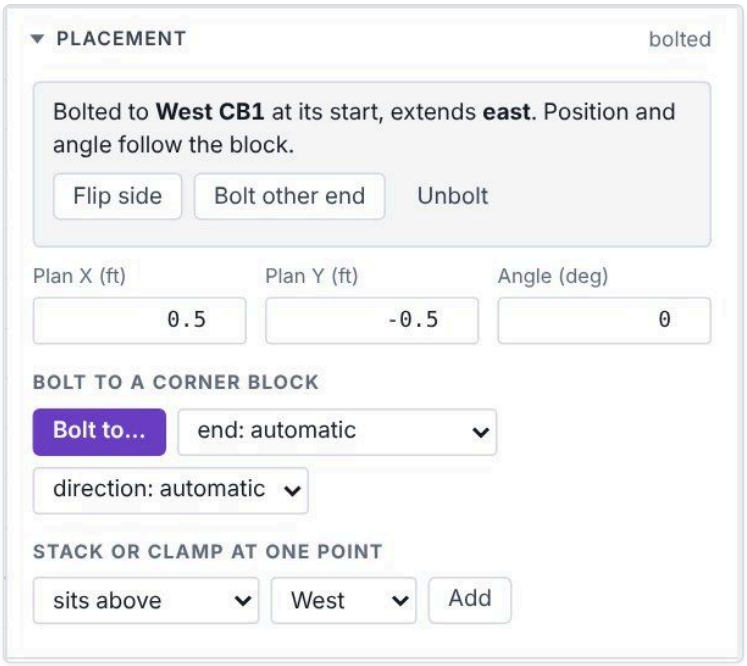
Choosing a corner block to bolt to. The blocks you can pick are outlined in blue.

You can also start from the block: select a corner block and click **Bolt a truss here...**, then click the truss; or click one of the block's free faces (**+ north**, **+ east**...) and then the truss, to bolt it to that face.

Once bolted, the **Placement** section says, for example, *Bolted to West CB1 at its start, extends east*, and offers:

- **Flip side** - swing the truss to the other side of the block's truss;
- **Bolt other end** - use the other end of this truss at the block;
- **Unbolt** - make it free again (it stays where it is).

Its X, Y and angle are now gray (set by the block). To move it, move the truss that owns the block.



The Placement section of a bolted truss.

Closing a Box: Bolting Both Ends

A truss can be bolted at both ends. After a truss is bolted at its start, bolt its **end** to a corner block on another truss: that other truss (if it is free) is **pulled onto** this truss so the block meets its end. Its panel then says *Pulled onto North (bolted through its own corner block)*, with **Flip side** and **Unbolt**. This is how the last side of a box closes onto the first - see Building a Box, Step by Step. If the lengths around a loop do not add up, the app warns that the truss does not meet the block and by how much.

Bolted Parts Must Match

Bolted truss and corner blocks only fit within one manufacturer's truss family (JTE 12×12 does not bolt to Christie A Type, even at the same size). The app warns, and draws the bolt in red on the plan, when bolted parts do not match - including a generic or universal truss bolted to branded parts. Parts it cannot check (generic, universal, custom) get a note. Trusses **stacked** or **clamped** at a crossing can be any family. Nothing is ever blocked; it is a warning.

8.7 Stacking or Clamping at a Crossing

Where two trusses cross on the plan, one can rest on top of the other or be clamped underneath it without a corner block. In **Placement > Stack or clamp at one point**, choose **sits above** or **clamped below**, choose the other truss, and click **Add**. The app adds a support where the two centerlines cross: this truss is then held up by the other truss at that point.

8.8 Bolted Connections Table

Every bolt and stacking point is a support of the truss, listed under **Hoists & connections > Bolted connections**. For each one you can see and change where it is on this truss (**At ... from**), which truss or block it is joined to and where on it, whether it is in the same plane, **above** or **below**, a hardware

weight at the connection (**+lb**), and the force passed through it. The **plan** tick keeps both distances tied to the plan drawing; typing a distance unticks it. **Delete** removes the connection.

8.9 UDL and Cable Allowance

- **UDL** - a total weight spread evenly along the whole truss, for example a drape, a projection screen or an LED wall hung along it. Type the total (not per foot). The Loads quick-add box also takes a UDL over part of the truss (Three Ways to Add a Load).
- **Cable** - cable weight per foot (lb/ft) on this truss, added to its UDL. Blank uses the rig's cable allowance (Rig Settings); `0` means no cable on this truss. The panel shows the result, e.g. *Cable allowance 1.5 lb/ft × 32 ft = 48 lb, added to the UDL.*

8.10 Advanced

Field	What it does
Height	Raises the truss in the 3D view only (no effect on results).
Width (in)	How wide it is drawn on the plan and in 3D. Taken from the truss type; type to override.
Stiffness (×)	Scales the truss's bending, shear and torsion stiffness in the whole-rig analysis (1 = the app's estimate). Only change this if you have better data.
Weightless	Leaves this truss's self weight out of the calculation. Rarely correct - use with care.

Below the fields the app lists the stiffness it used (EI, GA, GJ) and whether it was estimated from the tables or taken from section data.

8.11 Duplicating and Deleting

Duplicate makes a free copy called *... copy*, a little below and to the right of the original, with the same type, pieces, loads and hoists. The copy is not bolted to anything and has no corner blocks or bolted connections - add those again.

Delete (or `Delete` with the truss selected) removes the truss after asking. Its corner blocks go with it; trusses that were bolted to it or to its blocks stay where they are but are no longer bolted (and show *Not solved* if that was their only support), and hoists hung below it hang from the structure again. Undo brings it all back.

8.12 The Structure Grid

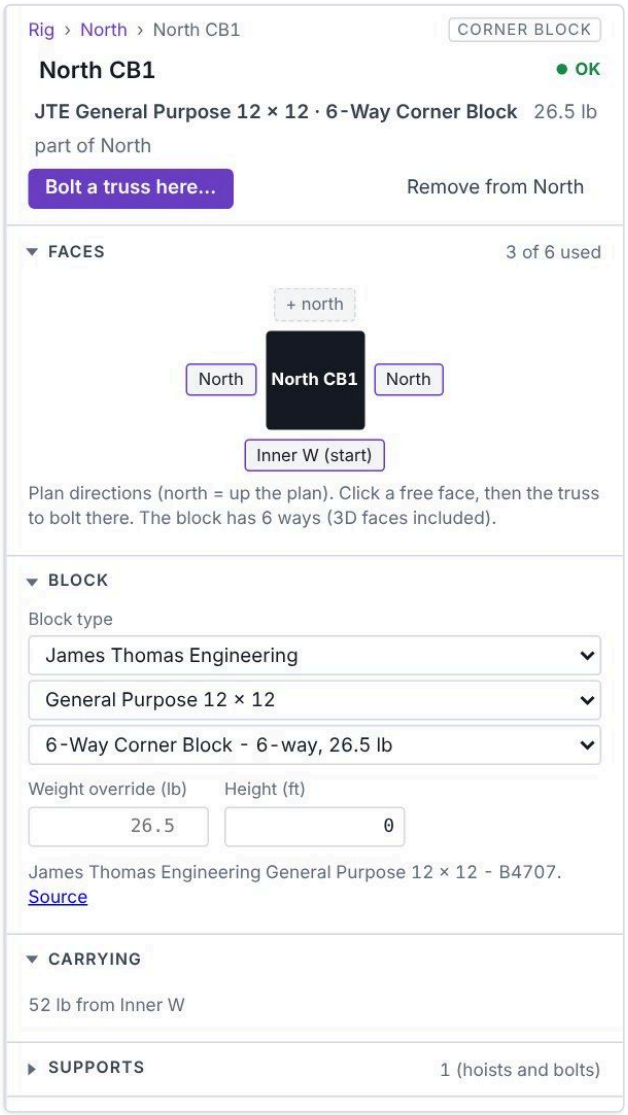
Column	Contents
Name	The truss name.
Type	Maker and model; type to search.
Line (pieces · blocks)	The line in words, e.g. `8' · CB1 · 13' · CB2 · 9'` (gray).
Pieces	Truss sections before corner blocks (gray when built from pieces or segments - change it in the panel).
Line	The whole line length (gray).
X, Y, Angle	Position of a free truss (gray when bolted).
Bolted to	The block it is bolted to and how (gray).
UDL	Spread load, total.
Measure from	start, CL or end.
Hangs from	What holds it up (gray).
Status	Its current status (gray).

8.13 Corner Block Panel

Click a corner block (on the plan, in the outline, or **select** in its truss's list) to open its panel.

- **Faces** - a diagram of the block with the truss on each face (north is up the plan) and how many of its ways are used. Click a free face, then a truss, to bolt that truss to that face. *Too many faces* means more trusses are bolted to it than it has ways.
- **Block** - the block type (maker > family > block); **Hardware fitted** for blocks sold with different plates; a **Weight override** (leave blank for the published weight - and enter one where the maker publishes none); its **Height** in 3D; and a link to the maker's product page.
- **Carrying** - the loads passed into it by the trusses bolted to it.
- **Supports** - a hoist can hang directly on a corner block: open Supports and click **+ Add hoist**.

Remove from ... takes the block out of its truss line.



A corner block's panel.

9 Building a Box, Step by Step

This example builds a simple box: two 22 ft sides (**West** and **East**, each 20 ft of truss with a corner block at each end) and two 30 ft sides (**North** and **South**) bolted between them, hung from four hoists, one at each corner. It uses the default truss (JTE 12×12 Plated) and the default 6-way corner block. (For a larger example, open **File > Load example box** and click each truss to see how it is built.)

9.1 The Two Short Sides

1. **File > New rig**. Click `+ Truss`. Rename it **West**. Set **Truss pieces** to `20` and **Angle** to `270` (it now runs south from X 0, Y 0).
2. Open **Corner blocks** and click `+ Corner block at start`, then `+ Corner block at end`. The line reads *20 ft truss + 2 ft corner blocks = 22 ft whole line*, and the blocks are *West CB1* (north end) and *West CB2* (south end).
3. Click `+ Truss` again. Rename it **East**, **Truss pieces** `20`, **Angle** `270`, **Plan X** `40` (anywhere to the east will do - it is pulled into place later). Add a corner block at its start and its end (*East CB1*, *East CB2*).

9.2 The Long Sides

1. Click `+ Truss`. Rename it **North**, **Truss pieces** `30`.
2. In **Placement**, set the options to **its start meets the block** and **extends east**. Click `Bolt to...` and click **West CB1** on the plan. North jumps to the north-west corner and runs east.
3. Set the option to **its end meets the block**, click `Bolt to...` again and click **East CB1**. East is pulled across so its north block meets North's end: the box has three sides.
4. Add **South** the same way: `+ Truss`, 30 ft; **its start meets the block**, **extends east**, `Bolt to...` **West CB2**; then **its end meets the block**, `Bolt to...` **East CB2**.

9.3 Loads and Hoists

1. Press `2`. Add some loads - for example select North, type `80 lb @ -5` in the quick-add box of its **Loads** section, press `Enter` and click **Mirror all**; do the same on South.
2. Press `3`. Put a hoist at each corner: on West and on East, add hoists at `0.5` from the **start** and `0.5` from the **end** (the centers of the corner blocks). In the grid, type the position in **At** and set **From** to *start* or *end*.
3. Press `4` and check the results. With 80 lb moving lights (two per side, mirrored) each corner hoist carries about 382 lb.



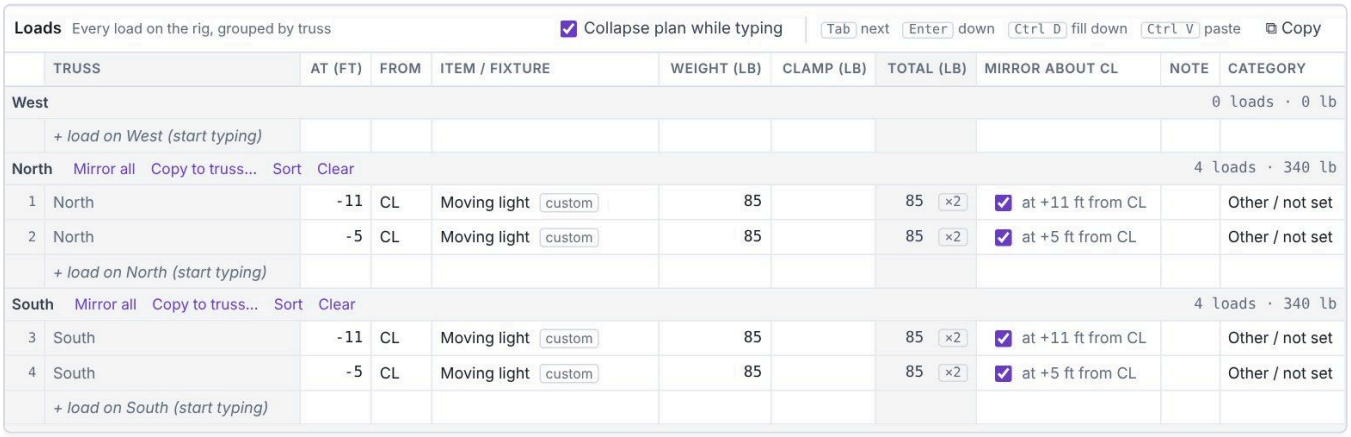
The finished box: four sides joined at four corner blocks, a hoist at each corner.

Why a hoist at every corner?

Some manufacturers (for example Christie Lites for its A Type) require a box built with 90° corner blocks to be supported at every corner, and allow only part of the truss capacity where an unsupported block joins four or more sections. The app warns about both. Always follow the maker's rules for its corner blocks.

10 Step 2 - Loads

In step 2 you put the loads on the trusses: fixtures from the library, weights you type, and spread loads. Every load is a point load at one position; for eight lights, add eight loads (or four, mirrored).



Loads Every load on the rig, grouped by truss — Collapse plan while typing — Tab next · Enter down · Ctrl D fill down · Ctrl V paste · Copy

	TRUSS	AT (FT)	FROM	ITEM / FIXTURE	WEIGHT (LB)	CLAMP (LB)	TOTAL (LB)	MIRROR ABOUT CL	NOTE	CATEGORY
West										0 loads · 0 lb
	+ load on West (start typing)									
North	Mirror all · Copy to truss... · Sort · Clear									4 loads · 340 lb
1	North	-11	CL	Moving light custom	85		85 ×2	at +11 ft from CL		Other / not set
2	North	-5	CL	Moving light custom	85		85 ×2	at +5 ft from CL		Other / not set
	+ load on North (start typing)									
South	Mirror all · Copy to truss... · Sort · Clear									4 loads · 340 lb
3	South	-11	CL	Moving light custom	85		85 ×2	at +11 ft from CL		Other / not set
4	South	-5	CL	Moving light custom	85		85 ×2	at +5 ft from CL		Other / not set
	+ load on South (start typing)									

The Loads grid: loads grouped by truss, each group ending with a "+ load on ..." row.

10.1 Three Ways to Add a Load

1. Quick Add, in the Truss Panel

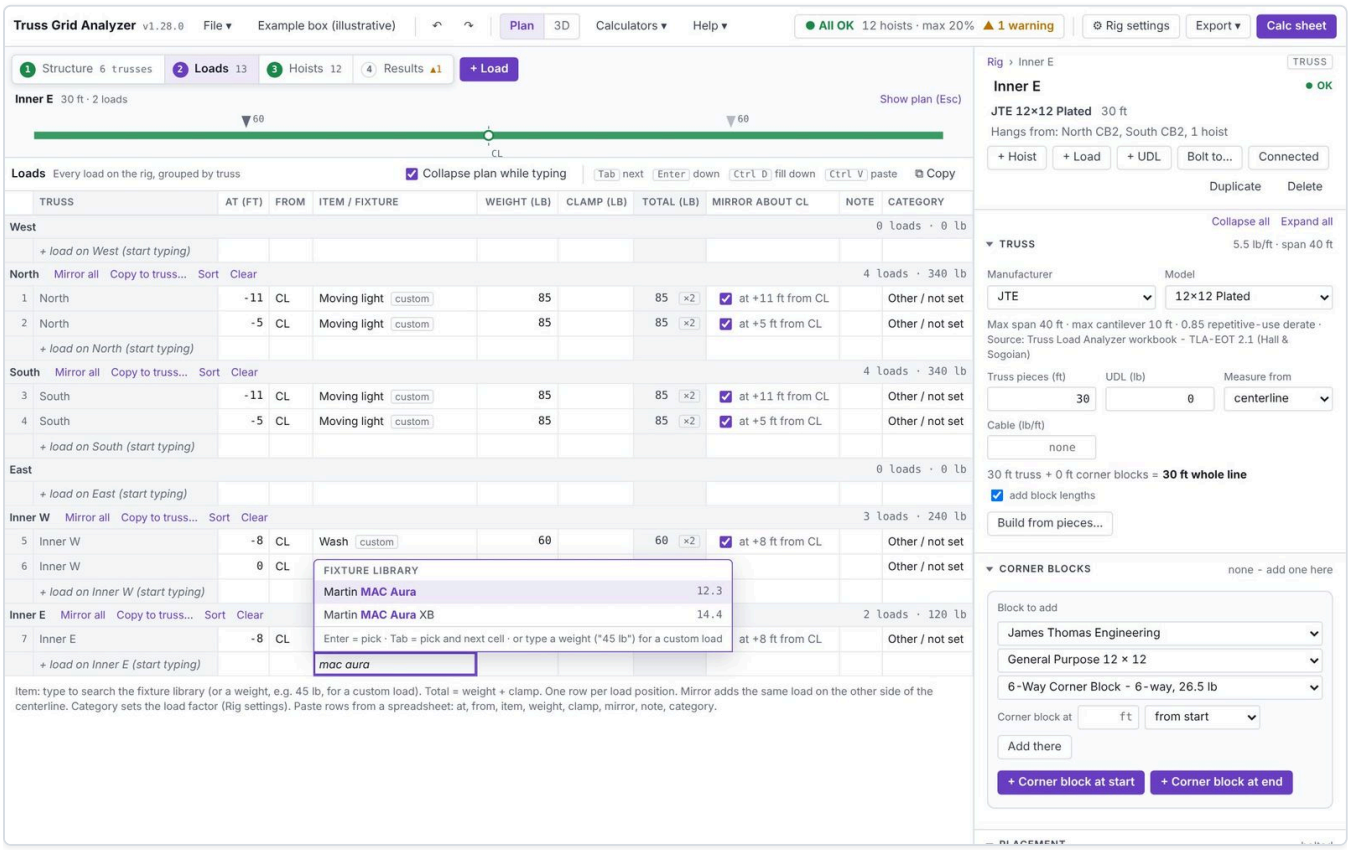
Select a truss and find the box *Add: fixture or weight* in its **Loads** section (or click **+ Load** at the top of the panel to jump there). Type what and where, and press `Enter`:

You type	You get
`MAC Aura @ 4`	A Martin MAC Aura (library weight and clamp) at 4 ft, measured the truss's way (from its centerline, start or end).
`120 lb @ -6`	A 120 lb load 6 ft before the centerline.
`20 kg @ 2.5`	A 20 kg load at 2.5 (meters in metric, feet in imperial).
`MAC Aura`	The fixture in the middle of the truss.
`UDL 200`	200 lb added to the truss's UDL, spread over its whole length.
`UDL 200 @ 2 to 10`	200 lb spread over the stretch from 2 to 10 ft (`2-10` also works). The solver's UDL always covers the whole truss, so the app enters this as equal point loads about a foot (0.3 m) apart, named after the UDL, and checks them as point loads against the CPL - conservative.
`UDL 25 lb/ft @ 2 to 10`	A weight per length (lb/ft or kg/m) instead of a total; here 25 × 8 ft = 200 lb. Without a range, per-length is added over the whole truss.

As you type a fixture name a list of matching fixtures appears; pick one, then type `@` and the position.

2. The Loads Grid

Click a cell in the *+ load on …* row of the truss's group and type. In the **Item / fixture** cell, type part of a fixture name to search the library - fixtures already used in this rig are listed first - or type a weight (`45 lb`) for a custom load.



Typing "mac aura" in a new load's Item cell. While you work in the grid the plan shrinks to a strip of the truss you are on.

3. The Add Button

With a truss selected in step 2, **+ Load** above the plan adds an empty load in the middle of it and opens it in the side panel. Fill in the item and weight there.

10.2 The Loads Grid Columns

Column	Contents
Truss	The truss it is on (gray).
At, From	Its position, and whether measured from the start, the centerline (CL) or the end.
Item / fixture	The fixture name, or your own description. `custom` means it is not a library fixture.
Weight	The fixture or item weight (each).
Clamp	The clamp weight, filled in from the library where it has one. Add your own.
Total	Weight + clamp (gray). `×2` means it is mirrored, so the truss carries it twice.
Mirror about CL	Tick to add the same load at the mirror position; the gray text says where (*at +11 ft from CL*).
Note	Anything you like (e.g. *SR 1/2*, *cable pick*). Printed on the calc sheet.
Category	Lighting, Audio, Video, Scenic, Rigging hardware or Other. Sets the load factor (see Allowances); the cell shows e.g. *Lighting ×1.1* when a factor applies. Library fixtures are Lighting.

Each truss group's header has **Mirror all** (mirrors every load on that truss), **Copy to truss...**, **Sort** (by position) and **Clear** (removes all its loads, after asking), plus the number of loads and their total weight.

10.3 Mirrored Loads

Rigs are often symmetrical. Tick **Mirror** on a load and the truss carries the same load at the same distance on the other side of its centerline. The twin is not a separate load: change the original's weight or position and the twin follows. A load exactly on the centerline has no twin. On the plan the twin's triangle is drawn lighter. Load counts (the Loads heading in the truss panel, the rig outline, the Loads grid and the step bar) count a mirrored load as two.

10.4 Copying Loads to Another Truss

Copy to truss... (in the grid group header or the truss panel) asks for the name of the truss to copy to, then copies every load keeping its distance from the **center** - so the pattern stays centered on a truss of a different length. Loads that would fall past the end of the other truss are placed at its end, and the app tells you how many.

10.5 Working with the Plan Strip

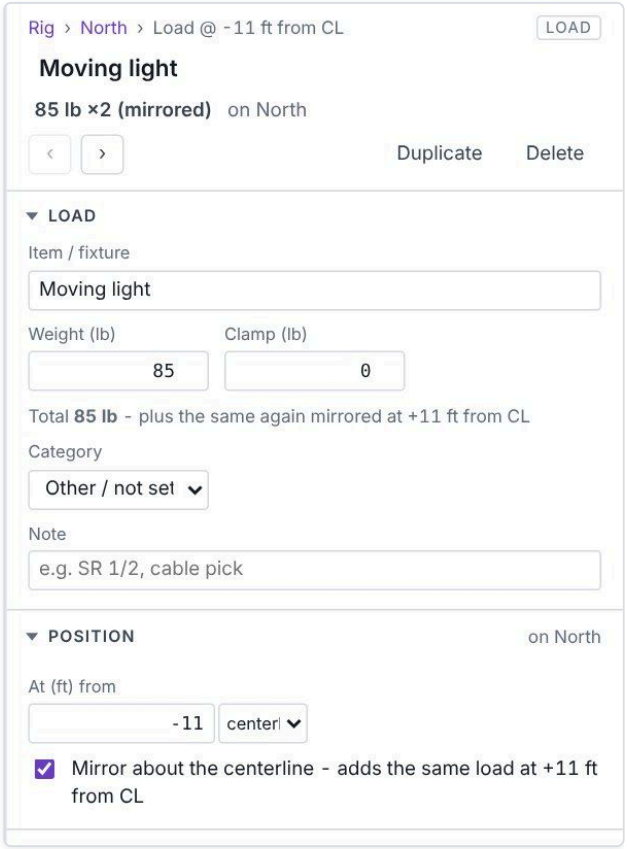
While a cell in the Loads grid is active, the plan shrinks to a strip showing just that truss: its centerline (CL), corner blocks, hoists and loads, with the load you are editing highlighted. Click **Show plan (Esc)** in the strip to bring the full plan back, or untick **Collapse plan while typing** in the grid header to keep the plan full size.

10.6 The Load Panel

Click a load in a truss panel's list or in the grid to open it in the side panel:

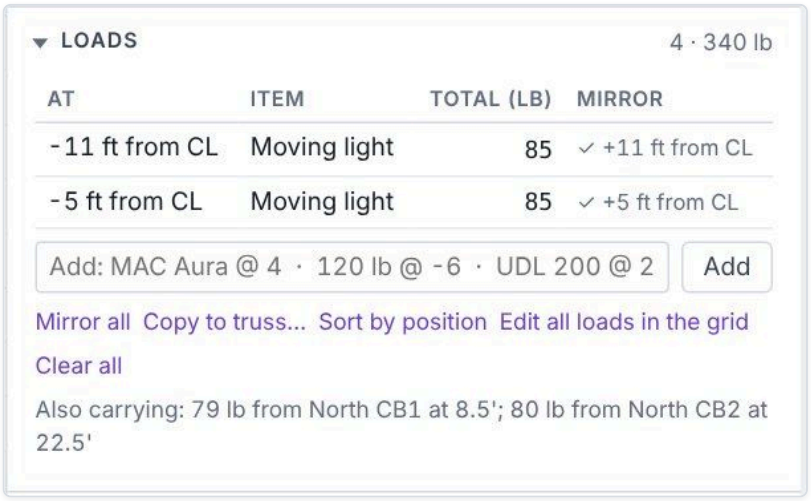
- **Item / fixture** - type to search the library; picking a fixture fills in its weight and clamp.
- **Weight** and **Clamp** - the total is shown underneath (and *plus the same again mirrored at …* when mirrored).
- **Category** - with a note of the load factor when one applies (e.g. *the trusses and hoists carry 93.5 lb*).
- **Note**.
- **Position** - **At … from** start / centerline / end, and **Mirror about the centerline**.

Use **‹ ›** to step through the loads on the truss, **Duplicate** to copy the load, and **Delete** (or `Delete`) to remove it.



A load's panel.

10.7 Loads in the Truss Panel



The Loads section of a truss.

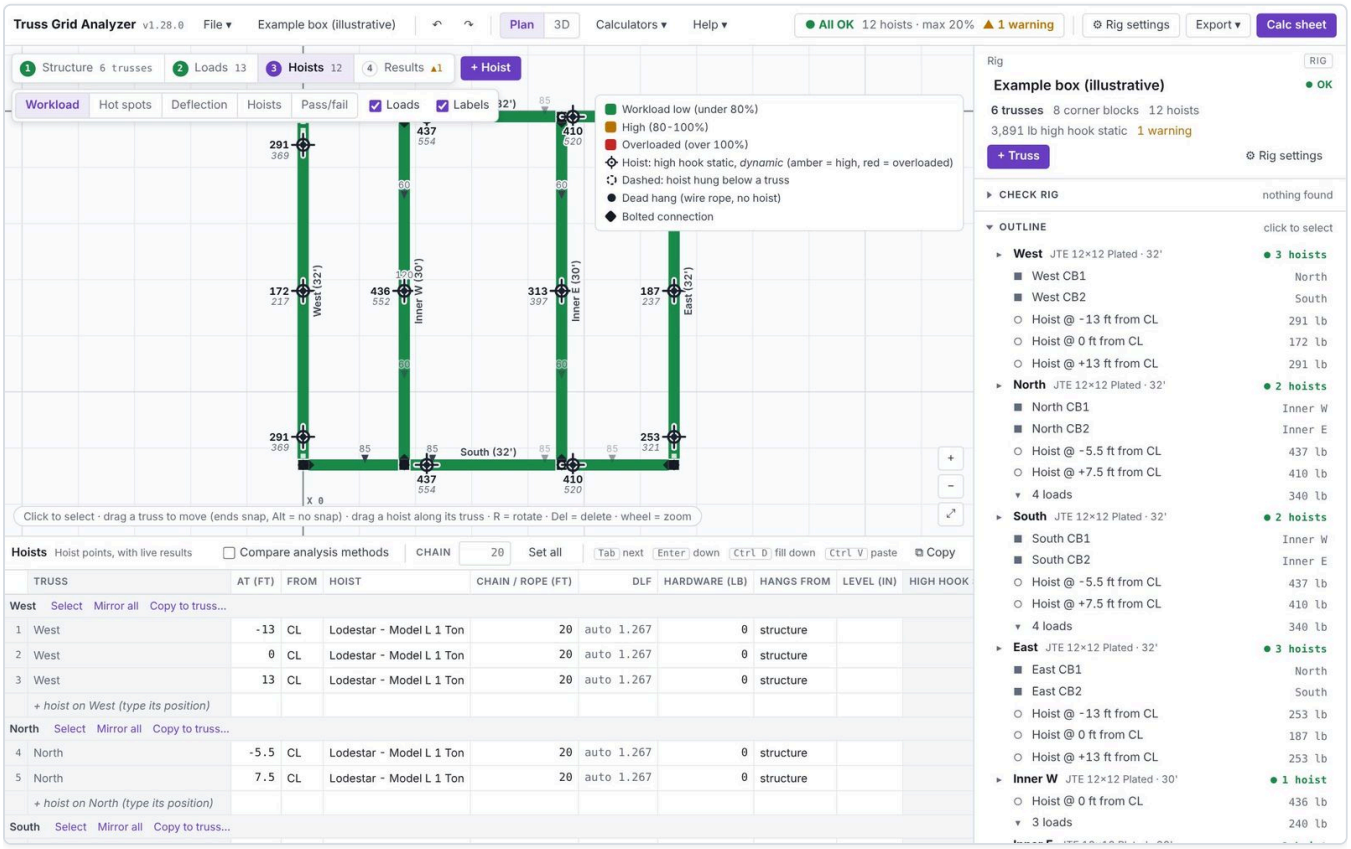
The truss panel's **Loads** section lists each load (click one to open it), with the quick-add box and the links **Mirror all**, **Copy to truss…**, **Sort by position**, **Edit all loads in the grid** and **Clear all**. The line *Also*

carrying lists loads that come into this truss from trusses bolted to it or its corner blocks, with where they land.

> **Get the weights right**
> A load with no weight is listed in **Check rig**. Remember cable, clamps, safeties, motors on moving trusses and anything else that will hang there: use a category load factor or the cable allowance for what you cannot list.

11 Step 3 - Hoists

In step 3 you place the hoists (and dead hangs), choose their models and chains, and see their loads live. A new truss has no hoists.



Step 3: each hoist on the plan shows its high hook static load and, in italics, its high hook dynamic load. The grid lists every hoist with its results.

11.1 Adding Hoists

- **In the grid**: click the **At** cell of the *+ hoist on …* row, type the position and press `Enter`. The cursor stays on the new row, so you can type the next position straight away.
- **With the button**: select the truss and click **+ Hoist** (above the plan, or at the top of the truss panel). The hoist is added in the middle of the truss; type its position in its panel or drag it.
- **On a corner block**: select the block, open **Supports** and click **+ Add hoist**.
- **By mirroring or copying** other hoists - see below.

A new hoist is set up like the **last hoist you placed** in the rig - same model, chain (or rope), hardware and dynamic factor - so a rig of identical hoists takes no extra typing. The very first hoist in a rig is a 1-ton CM Lodestar with 20 ft of chain.

11.2 Moving a Hoist

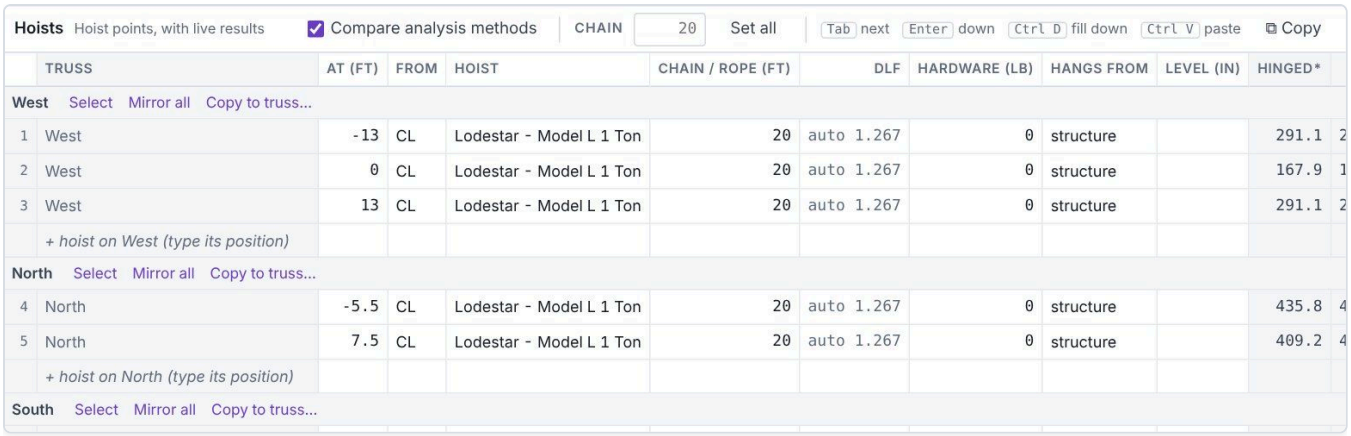
Type its position (**At** and **From** in the grid, or **Position** in its panel), or **drag it along its truss on the plan**. While you drag, the position is shown next to the hoist and the rig is re-solved as it moves. It moves in 1" steps (2 cm in metric), counted from the truss's measuring reference, so the distance stays a round number. Clicking a hoist without dragging just selects it.

11.3 The Hoists Grid Columns

Column	Contents
Truss, At, From	Where the hoist is.
Hoist	The model. Type to search: `lode 1t`, `prostar`, `2 ton`. Type `dead` for a dead hang.
Chain / rope	Chain length (or, for a dead hang, rope length). Its weight is added to the high hook load.
DLF	Dynamic load factor. Blank = automatic (shown gray, e.g. *auto 1.267*, from the hoist's speed); type a number to override.
Hardware	Weight of shackles, spansets, beam clamps and so on at this hoist.
Hangs from	*structure* (the usual), or *below* another truss - see Hoists Hung Below a Truss.
Level (in)	A designed level offset, + higher - see Hoist Levels.
High hook static / dyn.	Results (gray).
Workload, Status	High hook static ÷ capacity, and OK / Overloaded / Slack / Unstable.
Measured, vs calc.	A load-cell reading you type, and how far it is from the calculation - see Measured Loads.

The grid header also has:

- **Chain [ ] Set all** - type a length and click **Set all** to give every hoist that chain length (one Undo step).
- **Compare analysis methods** - adds three columns with each hoist's high hook static load from each joint model of the whole-rig analysis: **Hinged***, **Semi-rigid*** (a range) and **Rigid***. The high hook static is the largest of them.



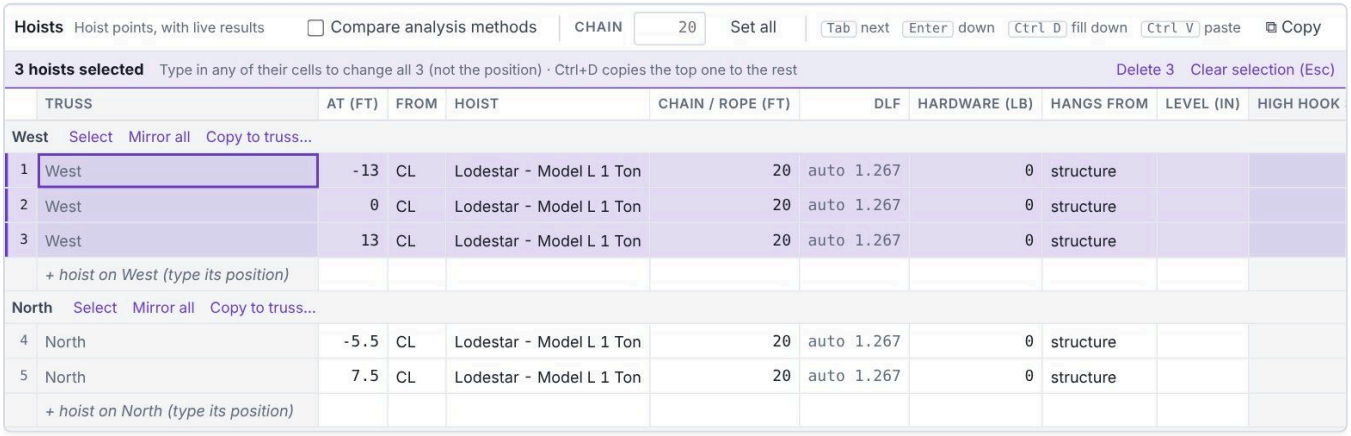
The Hoists grid with Compare analysis methods ticked.

Each truss's group header has **Select** (select all its hoists), **Mirror all** and **Copy to truss...**.

11.4 Editing Many Hoists at Once

In the Hoists grid you can select several hoists and change them all with one edit:

To select	Do this
A range	Click one hoist, then `Shift`+click another (on any cell or the row number).
One at a time	`Ctrl`+click hoists to add or remove them.
With the keyboard	`Shift`+`↓` / `Shift`+`↑` grows or shrinks the selection.
All hoists	`Ctrl`+`A` (with a grid cell active).
All hoists on one truss	Click **Select** in the truss's group header.



Three hoists selected. A bar above the grid says how many, with Delete and Clear selection.

Now type in any selected hoist's cell - model, chain, DLF, hardware, Hangs from, level, measured reading - and **every selected hoist** gets the same value. Positions (**At**) are never changed together. `Ctrl`+`D` copies the top selected hoist's value to the rest; `Delete` removes all of them (after asking). `Esc`, **Clear selection**, or moving with the arrows, `Tab` or `Enter` clears the selection.

11.5 Mirroring and Copying Hoists

- **Mirror** (in a hoist's panel) adds a hoist at the mirror position on the other side of the truss's centerline.
- **Mirror all** (truss panel, or the grid group header) does this for every hoist on the truss.
- **Copy to…** (hoist panel) and **Copy to truss…** (truss panel, grid) copy hoists to another truss - you type its name - keeping each one's distance from the center. Hoists past the end are placed at the end.

Unlike mirrored loads, each copy is a **real, separate hoist** you can edit on its own. It gets the same model, chain or rope, dynamic factor, hardware, level offset and Hangs from, but not the name or a load-cell reading. No copy is placed on the centerline or where a hoist already is; the app tells you what it left out.

11.6 The Hoist Panel

Click a hoist on the plan (or its row) to open its panel. At the top: its name (type one if you like, e.g. *DS-L*), its status, model, capacity and speed, **‹ ›** to step through the hoists on the truss, and **Mirror**, **Copy to...**, **Delete hoist**.

Result

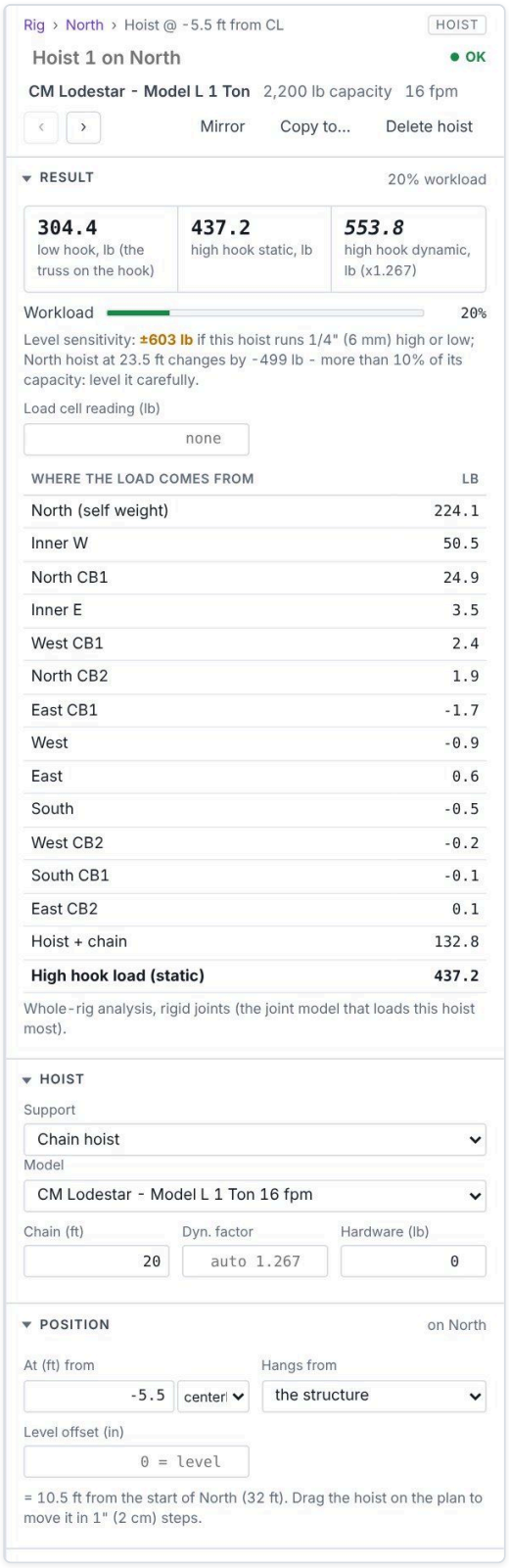
- Three figures: **low hook** (what the truss hangs on the hook), **high hook static** and **high hook dynamic** (with its factor).
- A **Workload** bar.
- **Level sensitivity**: *±603 lb if this hoist runs 1/4" (6 mm) high or low*, and which other hoist changes most. Highlighted when it is more than 10% of the hoist's capacity - level that hoist carefully.
- With an out-of-level tolerance set: how much it adds and whether the hoist could go slack.
- **Load cell reading** - type a measured value (see Measured Loads).
- **Where the load comes from** - every truss, corner block and load that puts weight on this hoist (its own truss's self weight, neighboring trusses through the grid - small negative figures mean a neighbor lifts a little off it), then Add %, hardware, hoist + chain, and the **high hook load (static)**. The note underneath says which joint model it came from.

Hoist

Support (Chain hoist or Dead hang), **Model**, **Chain** length, **Dyn. factor** (blank = automatic) and **Hardware**.

Position

At ... from, **Hangs from**, **Level offset**, and the distance from the start of the truss. Drag the hoist on the plan to move it.



A hoist's panel.

11.7 Hoist Models and the Dynamic Factor

The hoist list has 76 built-in chain hoists from the original workbook (CM, Chainmaster, Stagemaker, Prolyte and others) and any you add under Databases. Each has a capacity, a speed, a body weight and a chain weight per foot.

The **dynamic factor** allows for the extra load while a hoist starts, stops and runs. For a hoist with a listed speed it is **speed in ft/min ÷ 60 + 1** (16 fpm = 1.267, 32 fpm = 1.533). A hoist with no listed speed (for example a custom hoist) uses the rig's **Default dynamic factor** (1.25 unless you change it in Rig settings). You can type any factor for one hoist in its **DLF / Dyn. factor** box. A hoist's **status** compares its *static* high hook load with its capacity; the dynamic load is flagged separately if it exceeds capacity.

11.8 Dead Hangs

A **dead hang** is a support made of wire rope and hardware from the truss to the hang point, with no hoist. To make one, set the hoist's **Support** to **Dead hang (wire rope, no hoist)**, or type `dead` in the grid's Hoist cell. Then set:

Field	What to enter
Rope	The rope: 7×19 galvanized aircraft cable (1/8" to 3/8") or 6×19 IWRC EIPS (1/4" to 1/2"), each shown with its WLL. Or *not in the list - WLL typed below*. A new dead hang starts as 3/8" GAC.
Rope length	For its weight (and its stretch, if the rig's hoists are given a stiffness).
Assembly WLL	The working load limit of the weakest part of the assembly (a shackle, a fitting). It caps the rope's WLL.
Hardware	Shackles, beam clamp, spanset...
Dyn. factor	1.0 (static) unless you type one - a dead hang does not move.

The WLL is the rope's minimum breaking strength ÷ the **design factor** (10:1 by default; 7:1 or 8:1 in Rig settings), capped by the assembly WLL. The panel writes out the sum. Breaking strengths are typical catalog values - **check your rope's certificate**. A dead hang with no rope and no WLL is Overloaded. On the plan a dead hang is a filled dot; on the calc sheet it is DH1, DH2....

11.9 Hoists Hung Below a Truss

Sometimes a truss is flown from hoists that hang from another truss rather than from the building. Set such a hoist's **Hangs from** to **below** the upper truss (the *carrier*). Then:

- The hoist hooks onto the carrier where the plan puts it - so it must be directly under the carrier on the plan (the panel tells you where it hooks on, and warns if it is off to the side).
- Its chain joins the two trusses in the whole-rig analysis, so the carrier sags with the load and the truss below shares load with its other hoists. The chain can only pull.
- The carrier carries the hung hoist's high hook load, and the carrier (with everything bolted to it) is checked with the hung hoist's **high hook dynamic** load. So are the hoists holding the carrier up,

unless you turn that off in Rig Settings.

- On the plan the hung hoist is a **dashed** ring. The rig's total high hook load counts only hoists to the structure.

11.10 Hoist Levels

Two different things:

- **Level offset** (per hoist) - a *designed* difference: a truss hung raked or trimmed on purpose, or a dead hang whose length is fixed. Type inches (mm), + higher. The whole-rig analysis solves the rig with that point moved, and the loads shift accordingly.
- **Out-of-level tolerance** (Rig settings) - *unintended* error: any hoist may end up that far off (say ±1/4"). Every hoist's check then carries the worst it could see, and a hoist that could unload completely is warned about.

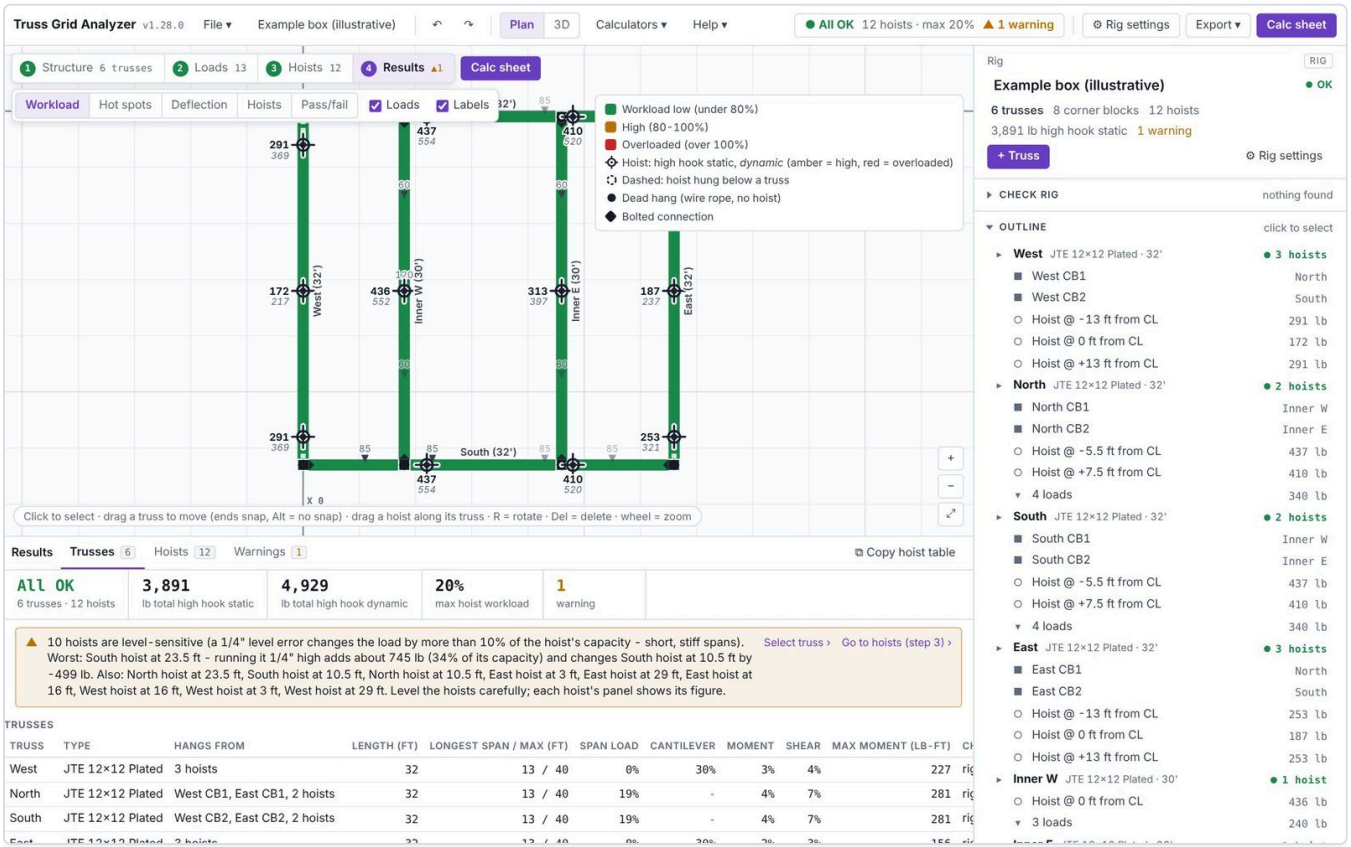
> **Level sensitivity matters on stiff rigs**
> On short, stiff spans, and in grids with many hoists, a small level error moves a lot of load. The app warns when a 1/4" error would change a hoist's load by more than 10% of its capacity. Level such hoists carefully, use load cells, or set an out-of-level tolerance so the checks include it. Giving hoists a stiffness (Rig settings) makes the analysis more realistic and much less level-sensitive.

11.11 Measured Loads (Load Cells)

If you have load cells, type each reading in the **Measured** column (or the hoist panel's **Load cell reading**). The app compares it with the calculated high hook static load - or with the low hook load if your cells hang between the hoist and the truss (set **Load cells read** in Rig settings). The **vs calc.** column shows the difference in percent. The rig panel's **Measured vs calculated** section totals each assembly of joined trusses and flags a total more than 5% off. One hoist on a rig that shares load can differ much more than that (levels, stiffness) - compare the assembly totals. Readings never change a result.

12 Step 4 - Results

Step 4 replaces the grid with the results. You can come here at any time; the results are always up to date.



Step 4 with the example box.

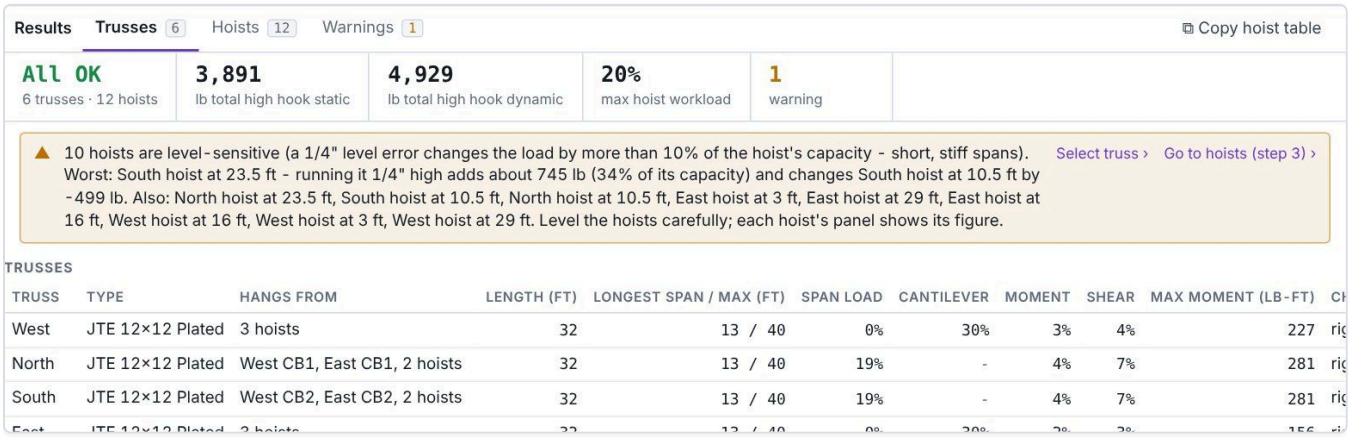
12.1 The Summary

Across the top of the results:

- **All OK** or **n to check** - the number of hoists and trusses that are not OK (overloaded, too long, slack, unstable or not solved);
- the **total high hook static** and **total high hook dynamic** load - what the whole rig puts on the structure (hoists to the structure only);
- the **maximum hoist workload**;
- the number of **warnings**.

Below it, the first warning is repeated as a banner, with links: **All n warnings**, **Select truss ›** and, for hoist problems, **Go to hoists (step 3) ›**.

12.2 Trusses Tab

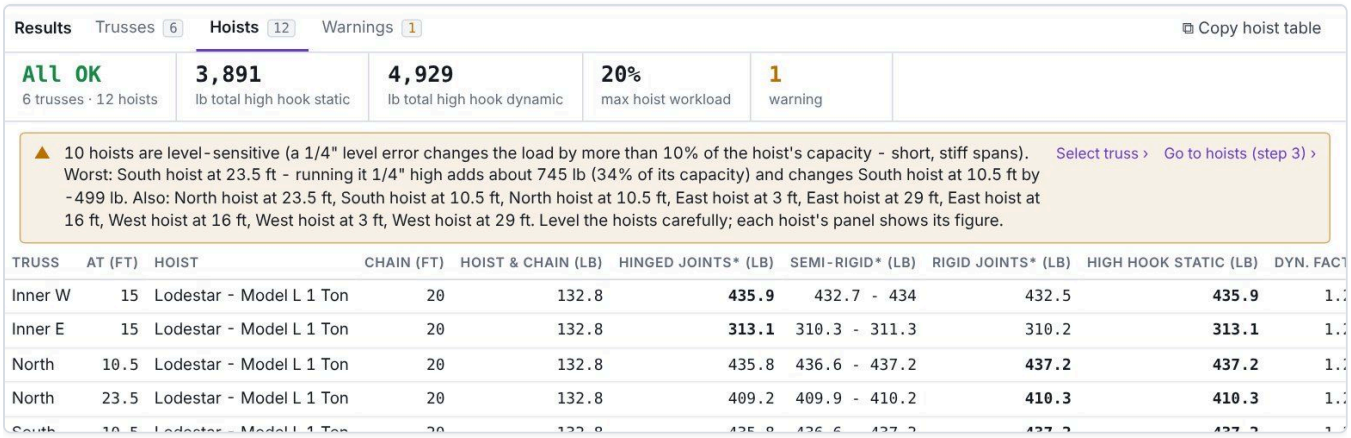


The Trusses tab.

Column	Meaning
Hangs from	What holds the truss up.
Longest span / max	Its longest span, and the longest the table allows (red if too long).
Span load	The worst span: its point loads as a share of the table capacity for that span (less any UDL).
Cantilever	The worst cantilever: its load against the table, or its length against a quarter of the maximum span, whichever is higher.
Moment, Shear	The largest bending moment and shear against the allowable estimated from the tables.
Max moment	The largest bending moment, lb-ft (kg·m).
Checked with	Which joint model of the whole-rig analysis loads this truss hardest (the one its checks use).
Status	OK, Too long, Overloaded, Too long and overloaded, Not solved.

Percentages of 80% or more are amber, over 100% red. The warnings list follows the table. Click a row to select the truss.

12.3 Hoists Tab



The Hoists tab.

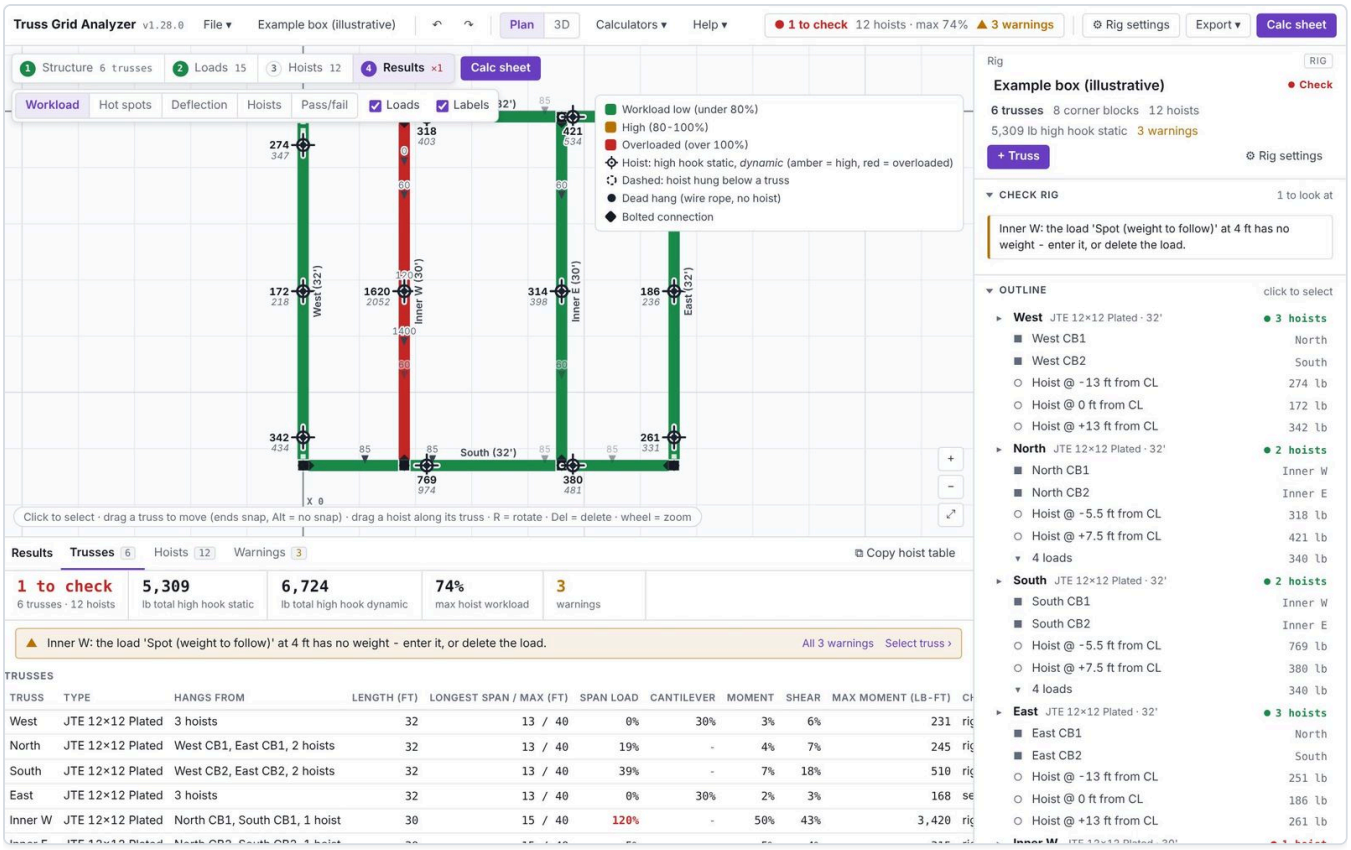
One row per hoist: truss, position (from the truss's start), model, chain, **Hoist & Chain** weight, the high hook static load from each joint model (**Hinged joints***, **Semi-rigid***, **Rigid joints***), the **High hook static** used (the largest, in bold), the dynamic factor, the **High hook dynamic** load, the capacity, the workload and the status. The note under the table explains the joint models. **⧉ Copy hoist table** copies the table for Excel (see Files, Export and Sharing).

12.4 Warnings Tab

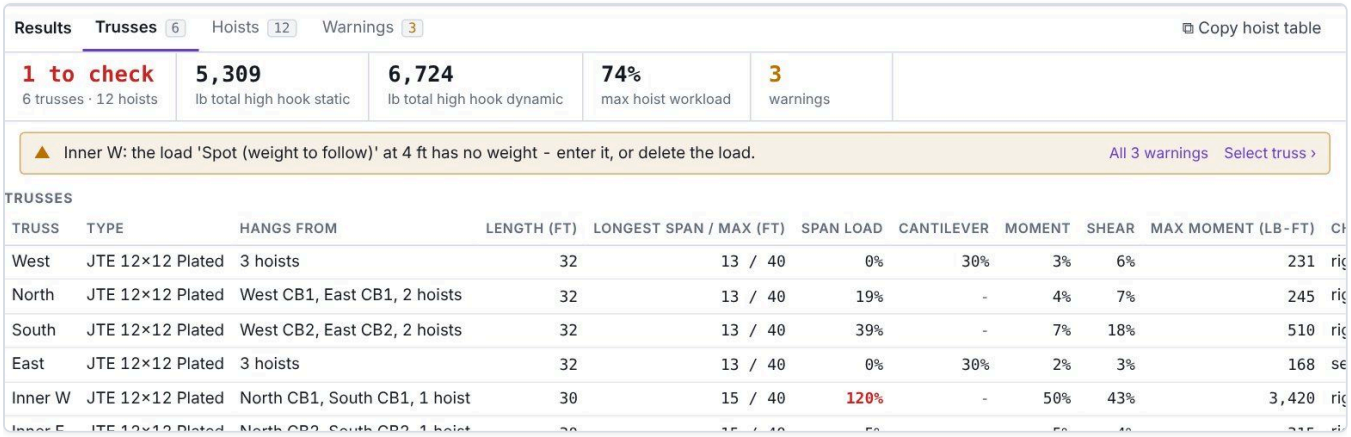
Every warning, then the **notes** (smaller, gray: things to be aware of that are not problems, such as parts the app cannot check). Click a warning to select the truss it is about. See Warnings and Messages for what each one means and what to do.

12.5 A Rig with Problems

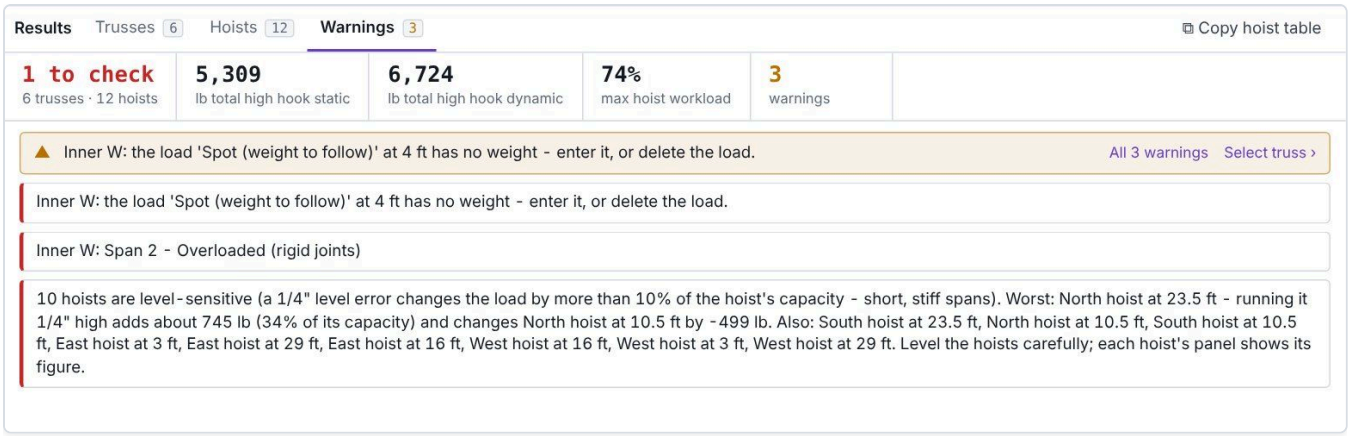
Here the example box has had a 1,400 lb video wall added to Inner W, 5 ft past its hoist, and a spot light entered with no weight:



Inner W is red on the plan; the status reads "1 to check" and "3 warnings"; Check rig lists the load with no weight.



The Trusses tab: Inner W's span load is 120% (red).

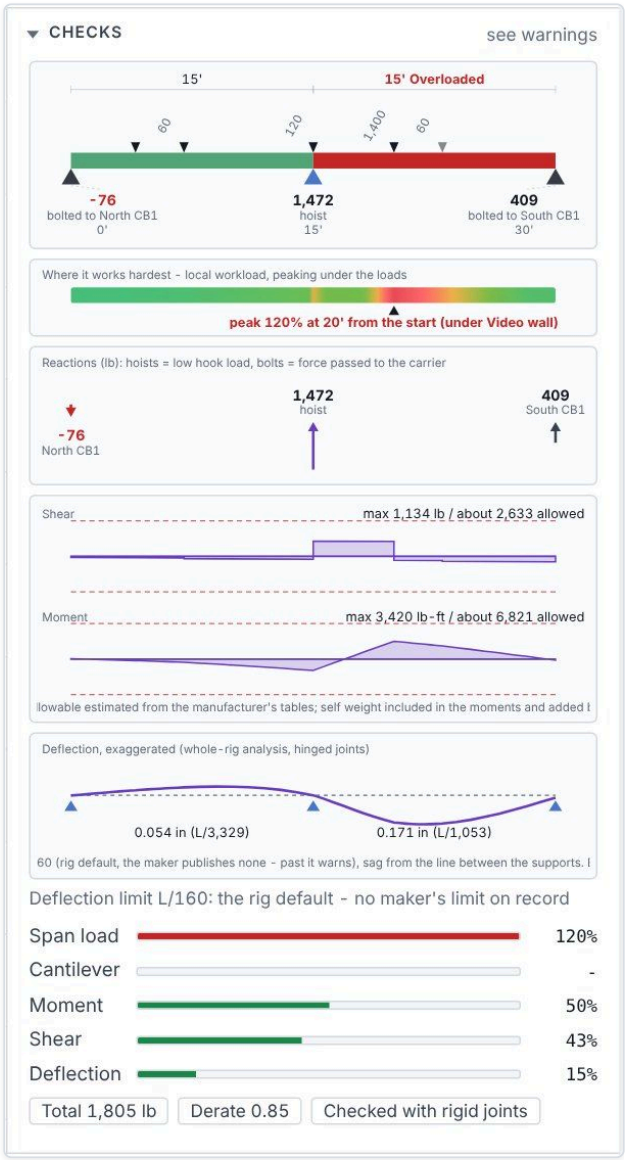


The Warnings tab explains each problem.

12.6 A Truss's Checks Section

Select a truss and open **Checks** in its panel to see exactly why it passes or fails. From the top:

1. **Elevation** - the truss from the side. Each span and cantilever is labeled with its length and colored (green OK, red failing, with the failure named). Loads are triangles with their weights (lighter: a mirrored twin; a different color: a load passed in from a bolted truss). Supports are triangles underneath with their reaction, what they are (*hoist* or *bolted to* ...) and their position.
2. **Where it works hardest** - a strip shaded by local workload (green through amber to red), with the peak marked and named.
3. **Reactions** - arrows to scale at each support. At hoists this is the low hook load; at bolts the force passed to the carrier. A red downward arrow means the truss is pushing up there.
4. **Shear and Moment** diagrams - with the estimated allowable as dashed lines and the maximum against it.
5. **Deflection** - the deflected shape, exaggerated, with each span's sag in inches and as span/ratio, and the limit used (the maker's, with a link to the data sheet, or the rig default).
6. **Bars** for Span load, Cantilever, Moment, Shear and Deflection.
7. **Chips**: the total load on the truss, the repetitive-use derate used, and *Checked with* ... joint



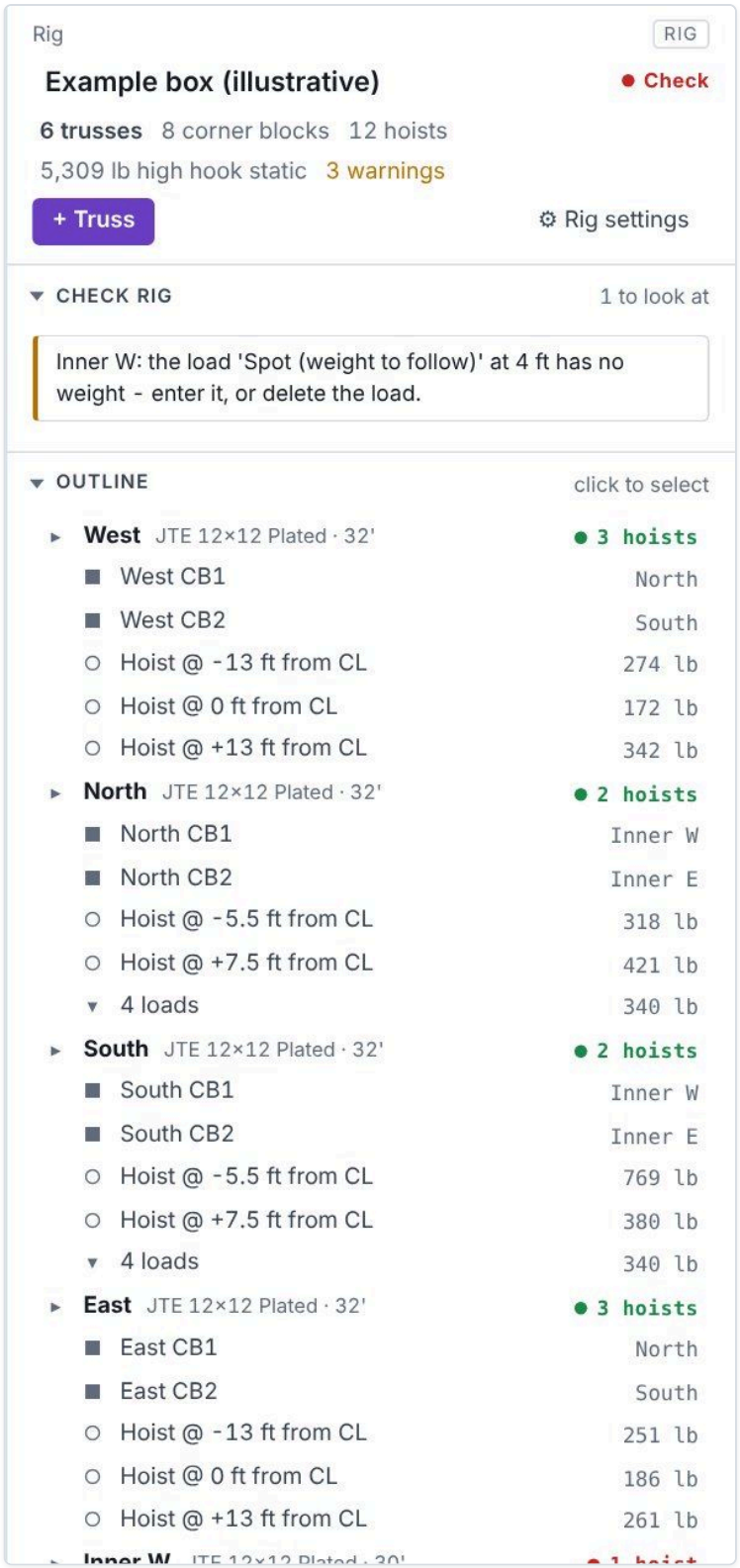
The Checks section of the overloaded truss.

model.

12.7 Check Rig

Click an empty part of the plan to show the rig panel. Its **Check rig** section lists input mistakes the calculation would otherwise take at face value:

- a load with no weight;
- two hoists at the same point (closer than 2") - is one a duplicate?;
- a bolted assembly with fewer than 3 hoists, or with all its hoists in one line (it could tip about that line);
- bolted ends that do not meet on the plan;
- trusses that cross on the plan with no joint between them (a note: they may be at different heights);
- a hoist with no model.



The rig panel with one Check rig item.

Click an item to select what it is about. If nothing is found it says so. Nothing in Check rig changes a result.

12.8 What to Do When Something Fails

Problem	Things to try
A hoist is Overloaded	Add a hoist nearby, move hoists to share the load, move or lighten loads, use a bigger hoist. Check the hoist's *Where the load comes from* table.
A span is Overloaded	Add a hoist in the span (shorter span, more capacity), move point loads toward supports, use a stronger truss.
A span or cantilever is Too long	Add a support or shorten the overhang (cantilever ≤ max span ÷ 4).
Moment or shear over	Usually a heavy load right next to a support, or bending over a support from a loaded neighboring span: move the load or add a hoist.
Deflection over	Shorten the span or lighten it.
A hoist is Slack	It would be pushed up - usually next to a heavy cantilever or span. Move it, or move the load; the other hoists carry more meanwhile.
Unstable	The truss or assembly has too few supports or they are in a line. Add hoists.
Not solved	It has no supports, or is bolted to something missing, or is part of a loop. See the warning.

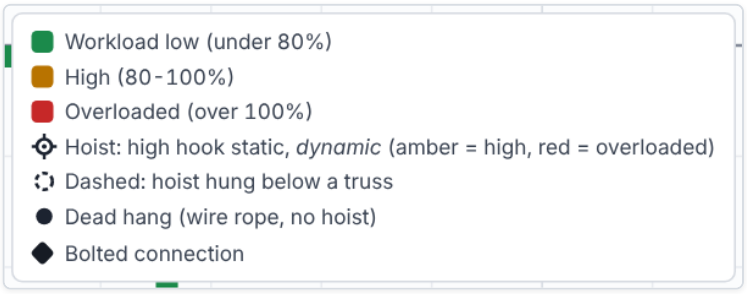
13 Plan and 3D Views

13.1 Moving Around the Plan

To	Do this
Zoom	Mouse wheel (zooms around the pointer), or the `+` / `−` buttons at the bottom right.
Pan	Drag an empty part of the plan.
See the whole rig	⤢ (bottom right) or press `F`.
Select	Click a truss, corner block or hoist. Click empty space to select the rig. The selected item glows.
Move a free truss	Drag it (half-foot steps; ends snap to other ends within 1.5 ft; hold `Alt` for no snapping). Bolted trusses and blocks cannot be dragged.
Rotate a free truss	`R` - 90° counterclockwise about its start.
Move a hoist	Drag it along its truss (1" / 2 cm steps).
Delete	`Delete` removes the selected load, hoist or truss (a truss asks first).
Get details	Hover over a truss, corner block, hoist or bolt for a tooltip (name, length, what it hangs from; block type, weight and ways used; hoist load, capacity and status).

The hint bar at the bottom of the plan is a reminder of these.

13.2 What the Plan Shows



The key (legend) in Workload mode.

- **Trusses** - bars to scale, colored by the current color mode, each labeled with its name and length (turned so it never reads upside down).
- **Corner blocks** - dark squares (red if more trusses are bolted to one than it has ways).
- **Bolted connections** - small diamonds; red when the bolted parts do not match (different truss families).
- **Hoists** - a ring with cross-hairs, labeled with the **high hook static** load and, in italics under it, the **high hook dynamic** load. The ring is amber when the hoist is high (or its dynamic load is over

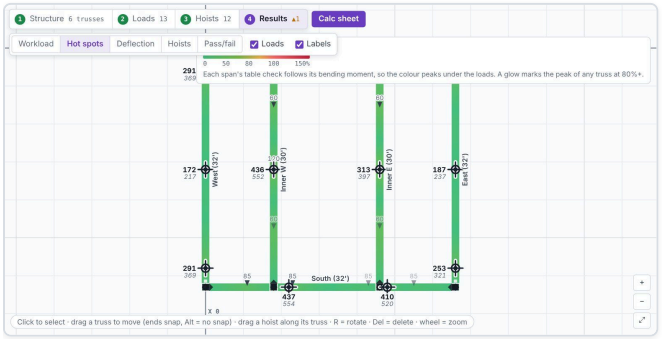
capacity) and red when it is overloaded or slack. A **dashed** ring is a hoist hung below a truss; a **filled dot** is a dead hang.

- **Loads** - small triangles with their weight (lighter for a mirrored twin). Untick **Loads** in the view toolbar to hide them.
- **Labels** - untick **Labels** to hide names and numbers for a cleaner picture.
- **X 0 / Y 0** - the origin lines.

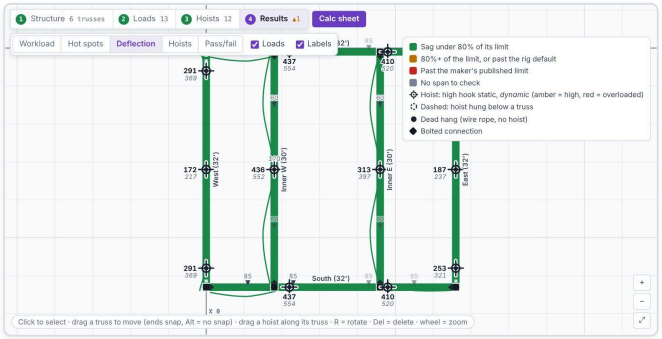
13.3 Color Modes

The buttons in the view toolbar choose what the truss colors mean. The key (bottom right of the plan) changes with them.

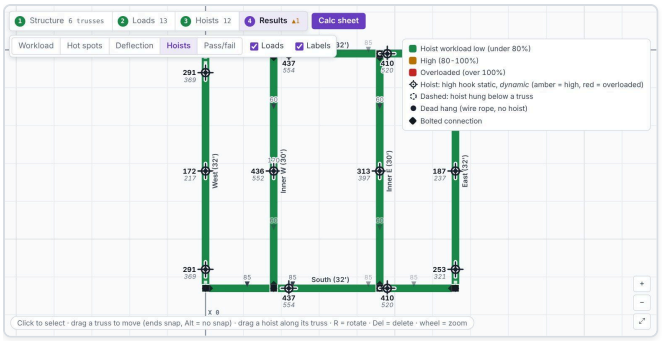
Mode	Colors each truss by...
Workload	Its highest workload - of its span, cantilever, moment, shear and deflection checks and the hoists on it: green under 80%, amber 80-100%, red over 100%. The default.
Hot spots	Local workload *along its length*, from green through amber to red, peaking under the loads that work it hardest. A glow marks the peak of any truss at 80% or more. Use it to find where on a truss the problem is.
Deflection	Its worst span's sag against its limit (green under 80%; amber from 80% or past the rig default; red past the maker's published limit; gray if it has no span to check). The deflected shape is drawn exaggerated beside each truss - a sag is drawn to its right-hand side. Hover a shape for its largest sag.
Hoists	The workload of the busiest hoist on it (gray: no hoists).
Pass/fail	Green if everything passes, red if anything fails.



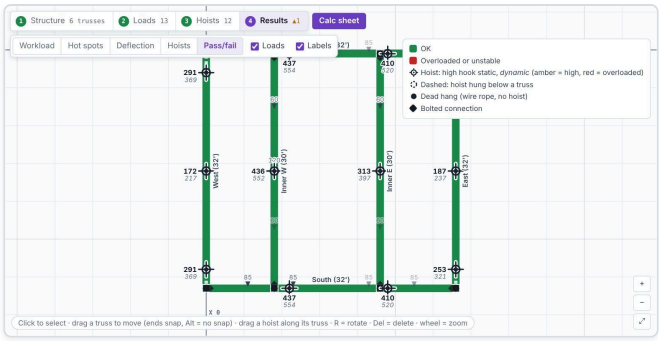
Hot spots.



Deflection: shapes drawn beside the trusses.



Hoists.



Pass/fail.

13.4 The 3D View

Click **3D** in the top bar to see the rig in 3D, with the hoists' chains rising to their hook points. It is for looking and presenting: you edit in the plan and the panels. Color modes work the same (in Deflection mode the shapes are drawn above the trusses).



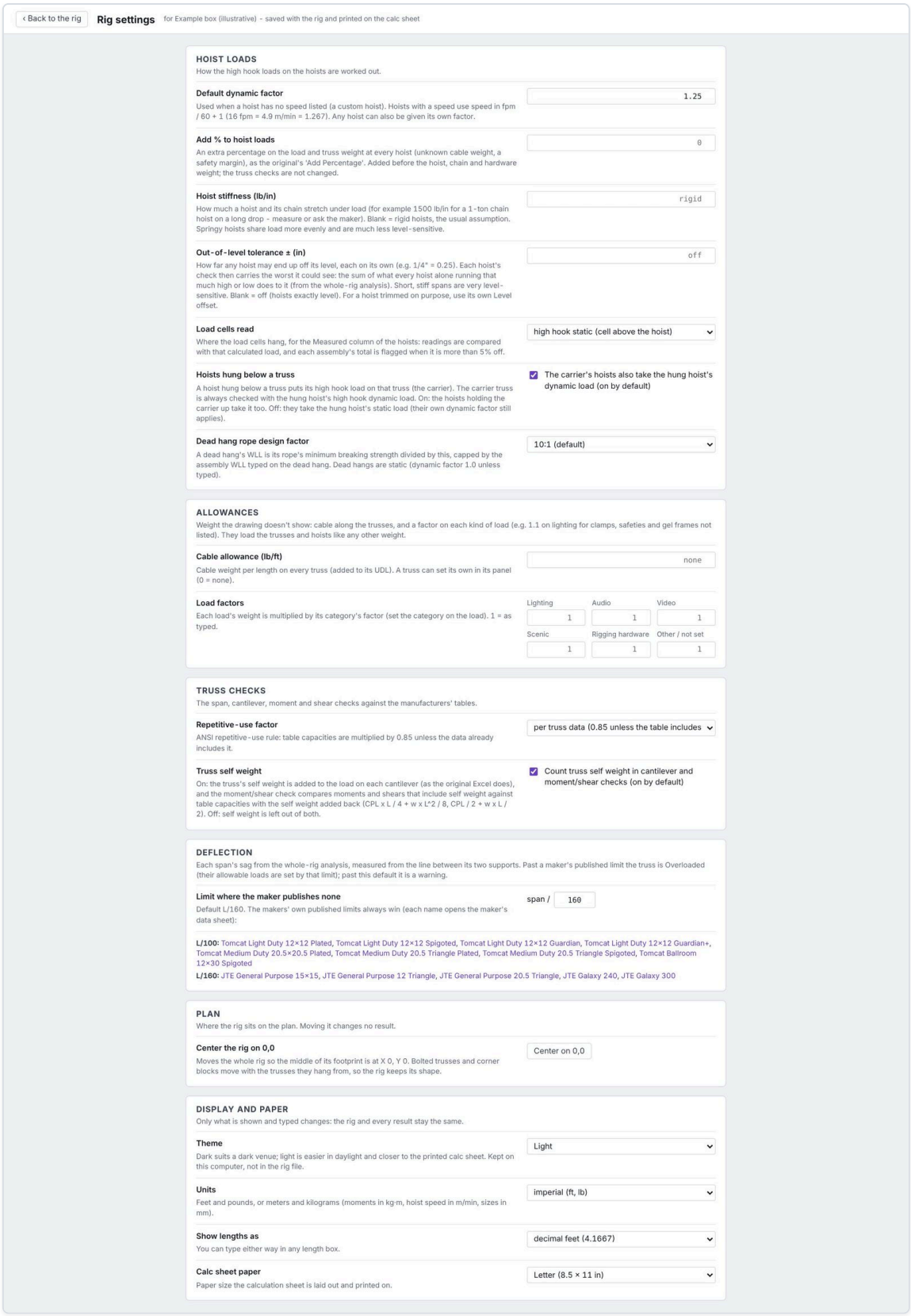
The 3D view.

To	Do this
Orbit	Drag an empty area, or use the **Rotate** and **Tilt** sliders.
Pan	`Shift` +drag, or right-drag.
Zoom	Mouse wheel. ⤢ or `F` fits the rig. (The + / − buttons zoom the plan only.)
Select	Click a truss or hoist; the side panel shows it as usual.

A truss's **Height** (Advanced section) raises it in 3D only, to separate trusses that are at different trims.

14 Rig Settings

Click **⚙ Rig settings** (top right, or in the rig panel) to open the settings for the whole rig. They are saved with the rig and printed on the calc sheet. Click **‹ Back to the rig** or press `Esc` to return.



The Rig settings page.

14.1 Hoist Loads

Setting	What it does	Default
Default dynamic factor	The dynamic factor for hoists with no listed speed (custom hoists). Hoists with a speed use speed in fpm ÷ 60 + 1; any hoist can have its own.	1.25
Add % to hoist loads	An extra percentage on the load and truss weight at every hoist - for unknown cable weight or a margin (the original workbook's "Add Percentage"). Added before the hoist, chain and hardware weight. It does not change the truss checks.	0 (off)
Hoist stiffness	How much a hoist and its chain stretch under load, lb/in (kg/mm) - e.g. 1500 lb/in for a 1-ton hoist on a long drop; measure it or ask the maker. Blank = rigid hoists, the usual assumption. Springy hoists share load more evenly and are much less level-sensitive.	rigid
Out-of-level tolerance ±	How far any hoist may end up off level, in inches (mm). Each hoist's check then includes the worst it could see. See Hoist Levels.	off
Load cells read	*high hook static* (cells above the hoists) or *low hook* (cells between hoist and truss) - what measured readings are compared with.	high hook
Hoists hung below a truss	On: the hoists holding a carrier truss also take a hung hoist's *dynamic* load. Off: they take its static load (with their own dynamic factor). The carrier truss itself is always checked with the dynamic load.	on
Dead hang rope design factor	7:1, 8:1 or 10:1. A dead hang's WLL = rope breaking strength ÷ this.	10:1

14.2 Allowances

Weight the drawing does not show. Allowances load the trusses and hoists like any other weight.

Setting	What it does
Cable allowance	Cable weight per foot (lb/ft, kg/m) on every truss, added to its UDL. A truss can have its own in its panel (0 = none).
Load factors	A multiplier for each load category - Lighting, Audio, Video, Scenic, Rigging hardware, Other / not set. For example 1.1 on Lighting to allow for clamps, safeties and color frames not listed. Set each load's category in its panel or the Loads grid. The loads keep the weights you typed; the calculation uses the factored weights.

14.3 Truss Checks

Setting	What it does	Default
Repetitive-use factor	The ANSI repetitive-use rule multiplies table capacities by 0.85 unless the maker's data already includes it. *per truss data* follows each truss's data (the generic Universal trusses use 0.75); or force *always 0.85* or *none (1.0)*.	per truss data
Truss self weight	On: each truss's own weight is added to the load on each cantilever, and included in the moment/shear check (and added back into the allowables, as the tables are loads on top of the truss's weight). Off: left out of both.	on

14.4 Deflection

Limit where the maker publishes none - span ÷ this number (default 160, i.e. L/160). Past this default a span gets a warning. Where a maker publishes a limit (Tomcat L/100; some JTE sheets L/160) that limit always applies, and past it the truss is Overloaded (the maker's allowable loads are set by it). The list under the setting names every truss with a published limit; each name opens the maker's data sheet.

14.5 Plan

Center on 0,0 moves the whole rig so the middle of its footprint is at X 0, Y 0. Bolted trusses and corner blocks move with the trusses they hang from. No result changes, and Undo moves it back.

14.6 Display and Paper

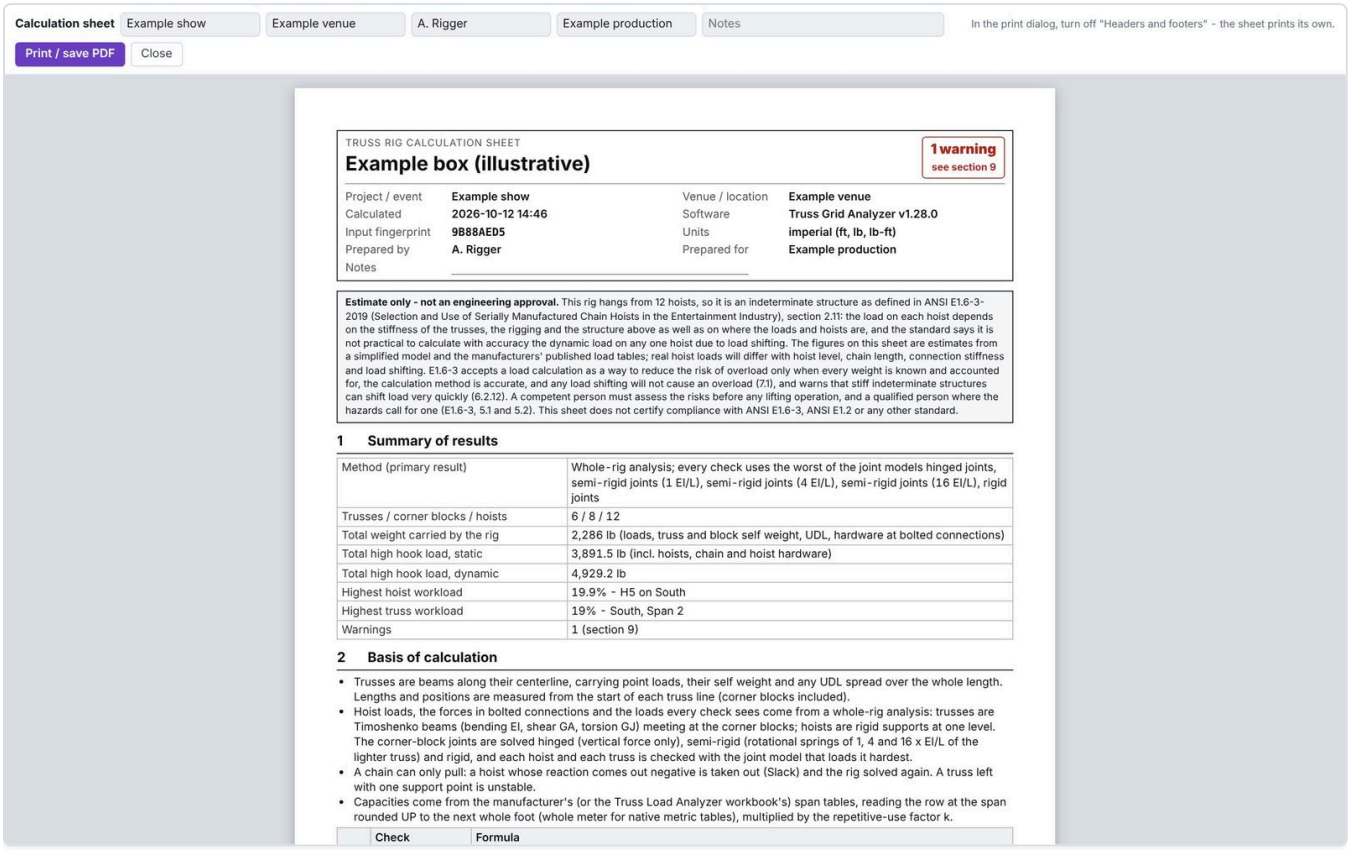
Setting	What it does
Theme	Dark, Light, or Follow the computer's setting. Kept on this computer, not in the rig file.
Units	imperial (ft, lb) or metric (m, kg; moments in kg·m, hoist speed in m/min, sizes in mm). Changes only what is shown and typed.
Show lengths as	(imperial) decimal feet (4.1667) or feet-inches (4'-2"). You can type either way in any box.
Calc sheet paper	Letter (8.5 × 11 in) or A4 (210 × 297 mm). Until you choose, Letter for imperial and A4 for metric.

15 The Calculation Sheet

The **calc sheet** is a printable record of the rig's calculation, written for independent review of the software's work: every input, formula, table look-up and result is set out so it can be checked by hand or in another program. It shows only what the app has calculated; opening it changes nothing.

15.1 Opening, Filling In and Printing

1. Click **Calc sheet** (top right, or the add button in step 4).
2. Fill in the boxes along the top: **Project / event**, **Venue / location**, **Prepared by**, **Prepared for** and **Notes**. They are printed on the sheet and saved with the rig.
3. Click **Print / save PDF**. In your browser's print dialog, choose a printer, or *Save as PDF* / *Microsoft Print to PDF* to make a PDF file. Choose the same paper size as the sheet (Rig settings > **Calc sheet paper**), and turn off the browser's own *Headers and footers* - the sheet prints its own.
4. Click **Close** or press `Esc` to go back to the rig. (While the sheet is open, the rig cannot be edited.)



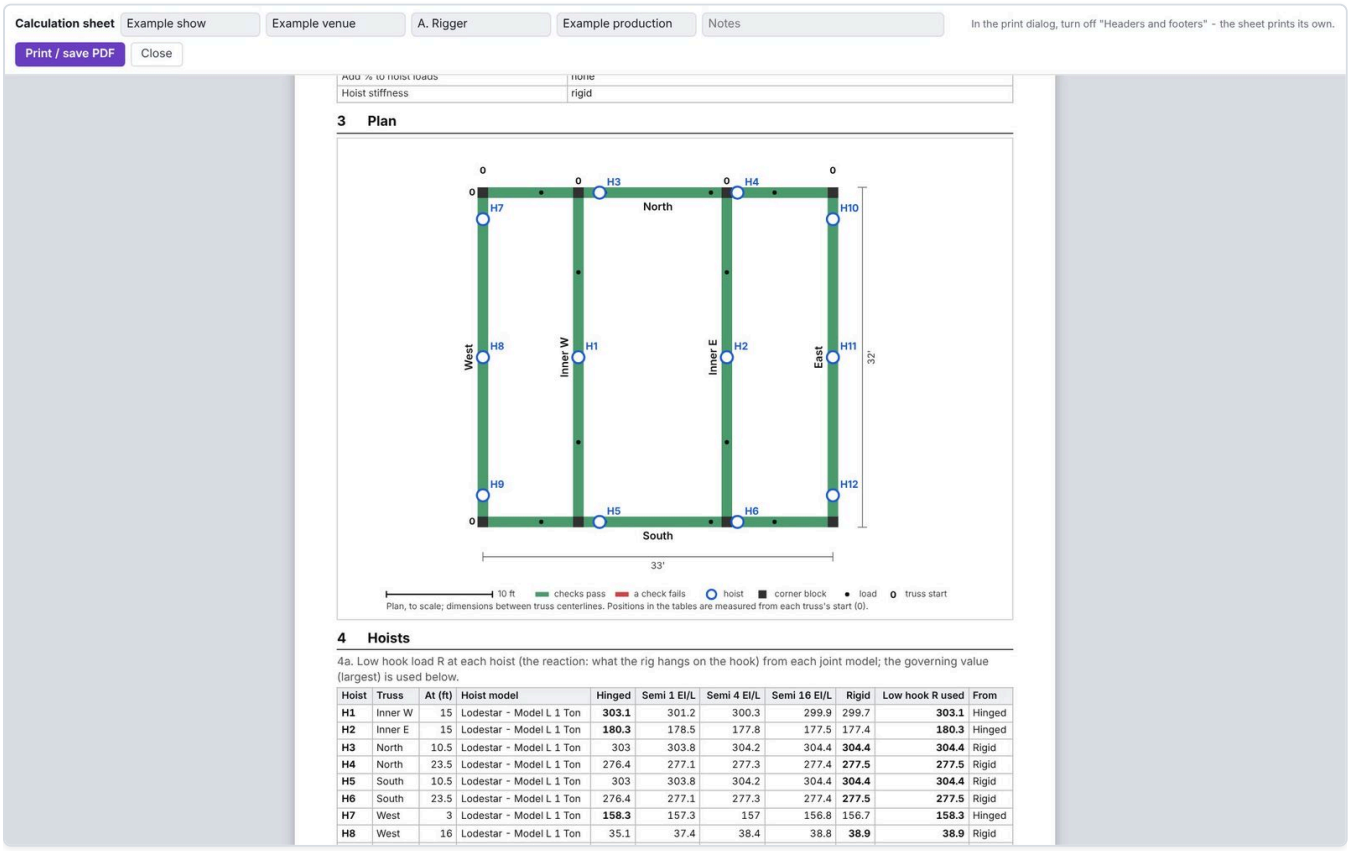
The top of the calc sheet: the title block and the first sections.

15.2 What Is on It

Part	Contents
Title block	Rig name, project, venue, the date and time calculated, the software version, units, notes, and an **input fingerprint** - a short code that changes whenever any input changes. Two sheets with the same fingerprint describe the same rig. A box at the top right shows the number of warnings or failures.
Estimate notice	Under the title block: the sheet is an estimate, not an engineering approval. On an indeterminate rig (one whose hoist loads depend on the stiffness of the trusses and rigging, such as most grids and boxes) it cites ANSI E1.6-3-2019 section 2.11, with 7.1, 6.2.12, 5.1 and 5.2, and says the sheet certifies no standard.
1 Summary of results	The method, counts, total weight carried, total high hook static and dynamic loads, the highest hoist and truss workloads, and the number of warnings.
2 Basis of calculation	The assumptions, and every formula used (F1 span, F2 cantilever, F3 moment and shear, ...) written with real fractions and symbols.
3 Plan	A plan drawn to scale with the hoists numbered H1, H2... (dead hangs DH1...).
4 Hoists	Each hoist's reaction from every joint model and its static and dynamic load worked out part by part.
5 Equilibrium	Self-checks: the loads going in balance the reactions coming out.
6 Trusses	For each truss: its data, the loads it carries (including from bolted trusses), reactions and balance, the diagrams, each span and cantilever check with the table row read and the arithmetic written out, the moment and shear check, and the deflection drawing (the exaggerated deflected shape) with the deflection check, formula F7: each span's sag against its limit L/n, where it is, L/d, workload and status, plus the cantilever tips (reported only).
7 Bolted connections and corner blocks	The forces through the bolts, and each corner block.
8 How to check this independently	Guidance for reviewing the results independently.
9 Warnings and notes	Everything from the Warnings tab.
Sign-off	Prepared by and Prepared for (also shown in the title block).

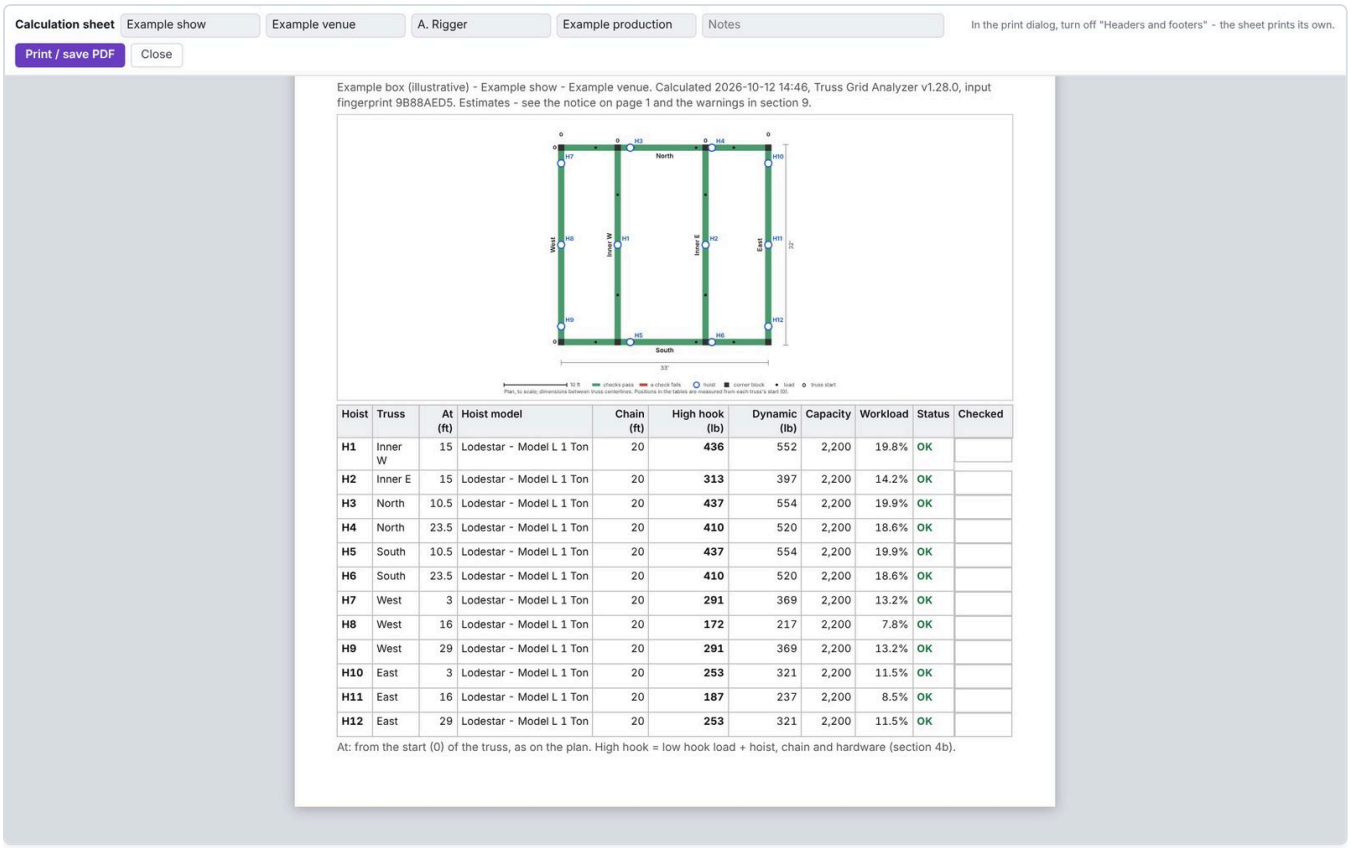
Part	Contents
A Hoist schedule	The last page, for the crew: the plan and one row per hoist - truss, position, hoist, chain, high hook static and dynamic, capacity, workload and a *Checked* column to tick off as each hoist is hung.

Every printed page has a footer with the rig name, the app version, the fingerprint, and *Page n of m*.



Hoist	Truss	At (ft)	Hoist model	Hinged	Semi 1 EI/L	Semi 4 EI/L	Semi 16 EI/L	Rigid	Low hook R used	From
H1	Inner W	15	Lodestar - Model L 1 Ton	**303.1**	301.2	300.3	299.9	299.7	**303.1**	Hinged
H2	Inner E	15	Lodestar - Model L 1 Ton	**180.3**	178.5	177.8	177.5	177.4	**180.3**	Hinged
H3	North	10.5	Lodestar - Model L 1 Ton	303	303.8	304.2	304.4	**304.4**	**304.4**	Rigid
H4	North	23.5	Lodestar - Model L 1 Ton	276.4	277.1	277.3	277.4	**277.5**	**277.5**	Rigid
H5	South	10.5	Lodestar - Model L 1 Ton	303	303.8	304.2	304.4	**304.4**	**304.4**	Rigid
H6	South	23.5	Lodestar - Model L 1 Ton	276.4	277.1	277.3	277.4	**277.5**	**277.5**	Rigid
H7	West	3	Lodestar - Model L 1 Ton	**158.3**	157.3	157	156.8	156.7	**158.3**	Hinged
H8	West	16	Lodestar - Model L 1 Ton	35.1	37.4	38.4	38.8	**38.9**	**38.9**	Rigid

Further into the sheet: checks written out as formula, numbers and result.



Hoist	Truss	At (ft)	Hoist model	Chain (ft)	High hook (lb)	Dynamic (lb)	Capacity	Workload	Status	Checked
H1	Inner W	15	Lodestar - Model L 1 Ton	20	**436**	552	2,200	19.8%	OK	
H2	Inner E	15	Lodestar - Model L 1 Ton	20	**313**	397	2,200	14.2%	OK	
H3	North	10.5	Lodestar - Model L 1 Ton	20	**437**	554	2,200	19.9%	OK	
H4	North	23.5	Lodestar - Model L 1 Ton	20	**410**	520	2,200	18.6%	OK	
H5	South	10.5	Lodestar - Model L 1 Ton	20	**437**	554	2,200	19.9%	OK	
H6	South	23.5	Lodestar - Model L 1 Ton	20	**410**	520	2,200	18.6%	OK	
H7	West	3	Lodestar - Model L 1 Ton	20	**291**	369	2,200	13.2%	OK	
H8	West	16	Lodestar - Model L 1 Ton	20	**172**	217	2,200	7.8%	OK	
H9	West	29	Lodestar - Model L 1 Ton	20	**291**	369	2,200	13.2%	OK	
H10	East	3	Lodestar - Model L 1 Ton	20	**253**	321	2,200	11.5%	OK	
H11	East	16	Lodestar - Model L 1 Ton	20	**187**	237	2,200	8.5%	OK	
H12	East	29	Lodestar - Model L 1 Ton	20	**253**	321	2,200	11.5%	OK	

At: from the start (0) of the truss, as on the plan. High hook = low hook load + hoist, chain and hardware (section 4b).

The hoist schedule on the last page.

16 Files, Export and Sharing

16.1 Saving a Rig as a File

File > Save downloads the rig as `*rig name*.rig.json` to your browser's download folder (your browser may ask where to save it). The file contains the whole rig - trusses, blocks, loads, hoists, rig settings, calc sheet fields - plus the custom trusses, corner blocks, hoists and fixtures you have added, and the version of the app that saved it. It is a small text file: back it up, email it, keep it with the show paperwork.

16.2 Opening a Rig File

File > Open... and choose a `.rig.json` file. It replaces the rig on screen (Undo brings that back until you close or reload the page; save it first to keep a copy). If the file cannot be read, the app says why and leaves the current rig alone.

> **Custom Trusses, Hoists and Fixtures in the File**
> Opening a file *adds* the custom trusses, corner blocks, hoists and fixtures saved in it to the ones already in this browser; nothing of yours is removed. An entry that is already here is reused. If the file's entry has the same id as a different entry here, it is given a new id and the opened rig is updated to match.

> **Files from other versions**
> Rig files record the version that made them. A newer version of the app opens older files. Results may differ between versions where the calculation has changed - the version history (**File > About**) says which versions changed results. Opening a file made by a *newer* version in an older one may not work.

16.3 Sharing a Rig

Send the `.rig.json` file. The other person opens it with **File > Open...** in their copy of the app. For people who do not have the app, send the calc sheet as a PDF, a picture of the plan, or the hoist table.

16.4 The Export Menu

Item	What it does
Copy hoist table	Copies every hoist's loads to the clipboard as CSV text. Paste into Excel or Google Sheets. Columns: Truss, At, Low hook, Hoist & Chain, Added %, the high hook load from each joint model (hinged, semi-rigid min and max, rigid), High hook static, Governing (joint model), Dynamic factor, High hook dynamic, Capacity, Status. The same button is at the top of step 4.
Save view as PNG	Downloads a picture of the plan (or the 3D view, if showing) at twice screen resolution, named after the rig. Set up the color mode, labels and zoom first.
Export whole-rig model	Downloads the rig's whole-rig analysis model with the app's results (`.grillage.json`), for an engineer to cross-check in PyNite (the 3D frame engine behind CalcForge) using the project's `tools/pynite_check.py` . Most users will not need this.
Print screen	Prints the current screen through the browser. For a proper record, use the calc sheet.

Each grid also has a **⧉ Copy** button that copies the whole grid (tab-separated) for a spreadsheet.

16.5 Using the App on Another Computer

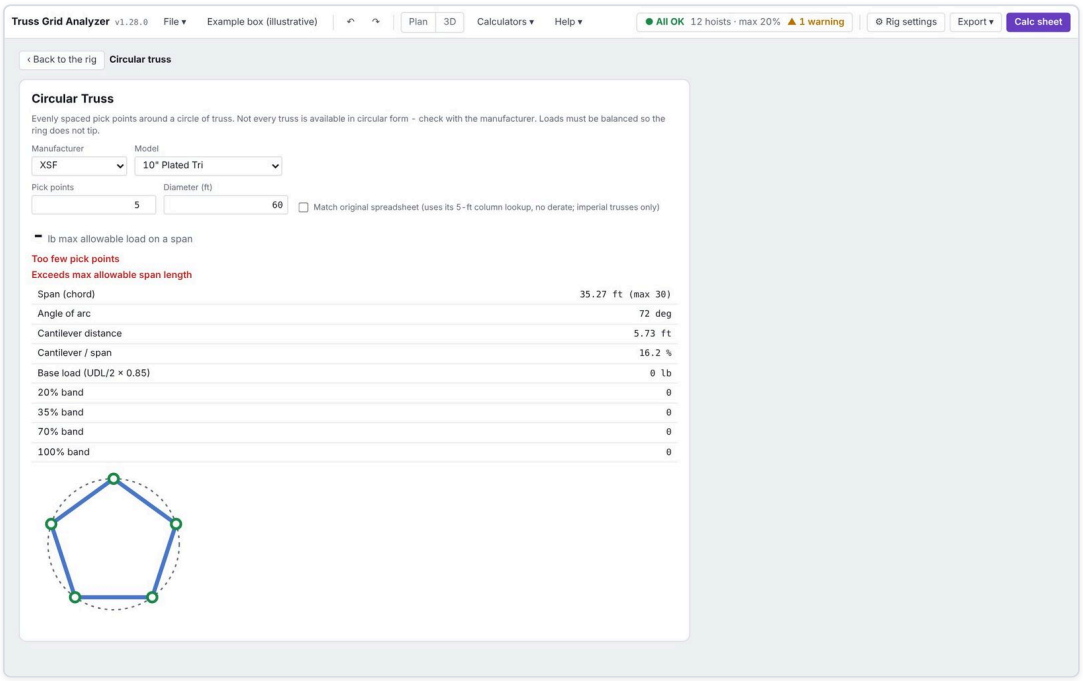
Copy the app file and your `.rig.json` files. Custom trusses, corner blocks, hoists and fixtures travel inside every rig file you save, so opening your most recent rig file on the new computer brings them along.

17 Calculators and Databases

The **Calculators** menu opens standalone tools in place of the rig. Click **‹ Back to the rig** to return. They do not change the rig.

17.1 Circular Truss

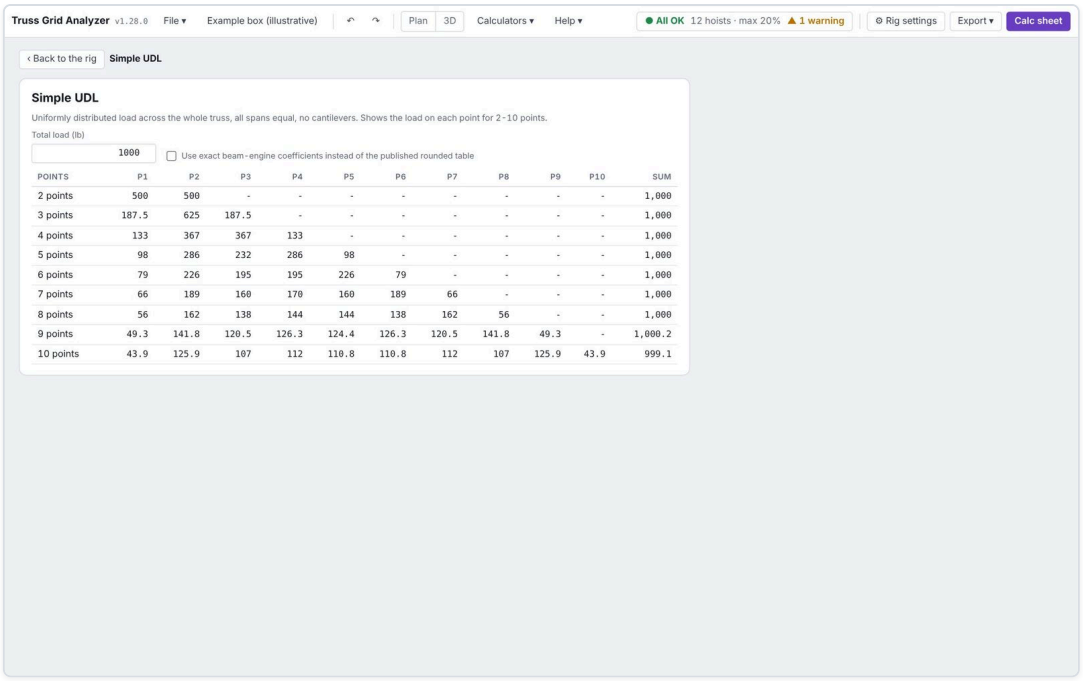
For a ring of truss hung from evenly spaced pick points. Choose the truss (manufacturer, model), the number of **pick points** and the **diameter**. It shows the **maximum allowable load on a span** (between two picks), the span (chord) against the maximum span, the angle of arc, the cantilever distance (how far the arc bows out from the straight chord) and its ratio to the span, the base load and the 20%, 35%, 70% and 100% load bands, with a drawing. **Match original spreadsheet** reproduces the original workbook's method (its 5-ft column look-up, no derate; imperial trusses only). Not every truss is made in circular form - check with the manufacturer - and loads must be balanced so the ring does not tip.



The Circular truss calculator.

17.2 Simple UDL

A quick table for a uniformly distributed load across a truss on 2 to 10 equally spaced points with no cantilevers: type the total load and read the load on each point. By default it uses the published rounded coefficients; tick **Use exact beam-engine coefficients** for the exact values.

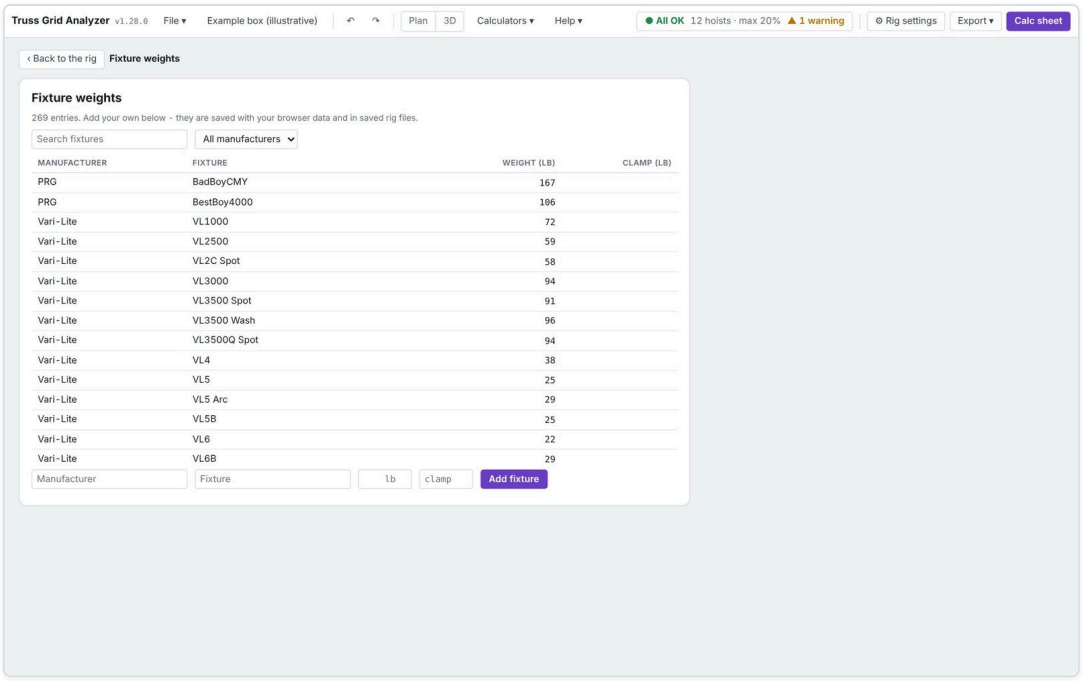


POINTS	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	SUM
2 points	500	500	-	-	-	-	-	-	-	-	1,000
3 points	187.5	625	187.5	-	-	-	-	-	-	-	1,000
4 points	133	367	367	133	-	-	-	-	-	-	1,000
5 points	98	286	232	286	98	-	-	-	-	-	1,000
6 points	79	226	195	195	226	79	-	-	-	-	1,000
7 points	66	189	160	170	160	189	66	-	-	-	1,000
8 points	56	162	138	144	144	138	162	56	-	-	1,000
9 points	49.3	141.8	120.5	126.3	124.4	126.3	120.5	141.8	49.3	-	1,000.2
10 points	43.9	125.9	107	112	110.8	110.8	112	107	125.9	43.9	999.1

The Simple UDL table.

17.3 Fixture Weights

The fixture library: 269 fixtures from the original workbook, with weights and (where known) clamp weights. Search by name, or filter by manufacturer. To add your own fixture, fill in **Manufacturer**, **Fixture**, weight and (optionally) clamp weight at the bottom and click **Add fixture**. Your fixtures appear in every fixture search, are kept in this browser, and are saved inside your rig files.

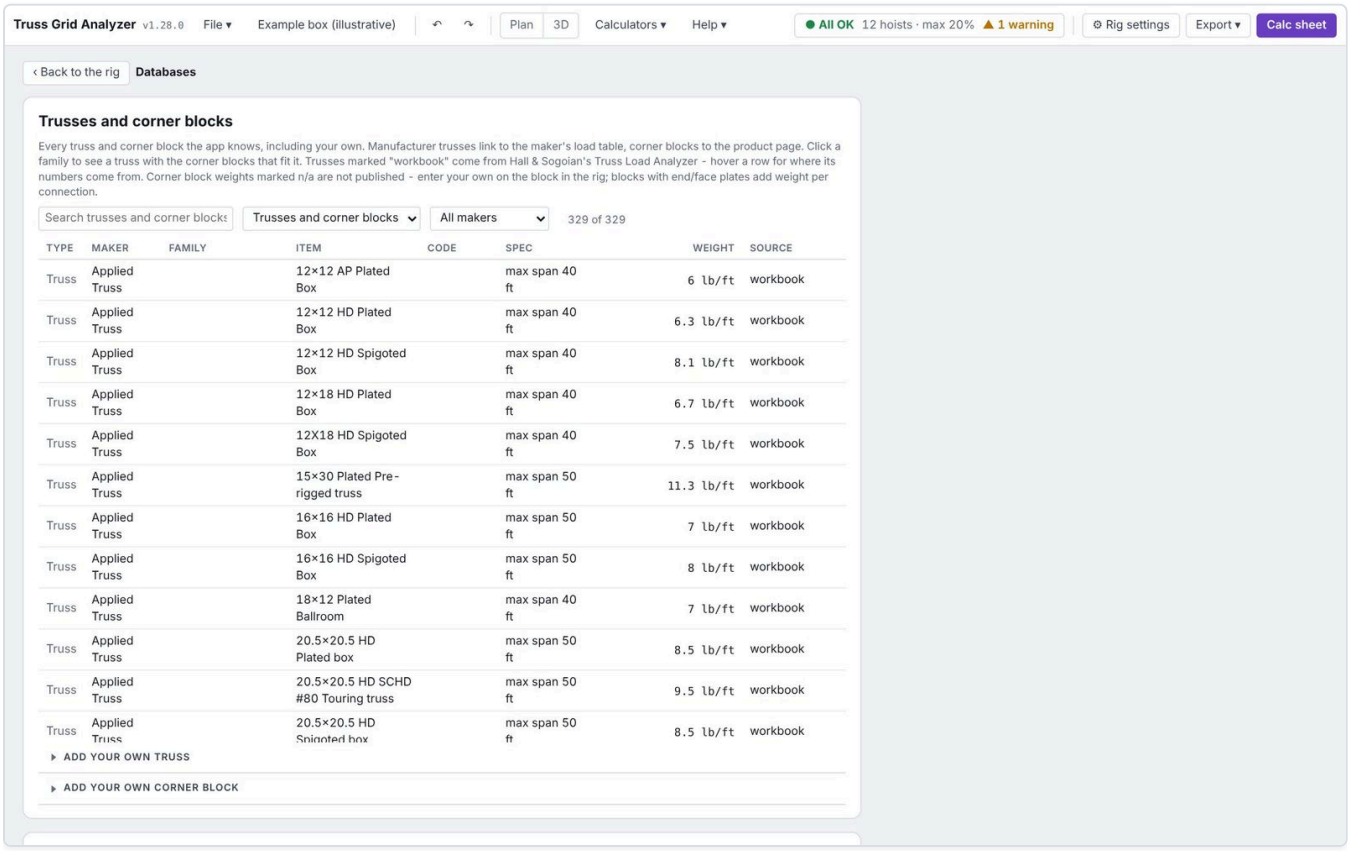


MANUFACTURER	FIXTURE	WEIGHT (LB)	CLAMP (LB)
PRG	BadBoyCMY	167	
PRG	BestBoy4000	106	
Vari-Lite	VL1000	72	
Vari-Lite	VL2500	59	
Vari-Lite	VL2C Spot	58	
Vari-Lite	VL3000	94	
Vari-Lite	VL3500 Spot	91	
Vari-Lite	VL3500 Wash	96	
Vari-Lite	VL3500Q Spot	94	
Vari-Lite	VL4	38	
Vari-Lite	VL5	25	
Vari-Lite	VL5 Arc	29	
Vari-Lite	VL5B	25	
Vari-Lite	VL6	22	
Vari-Lite	VL6B	29	

The fixture library.

17.4 Databases

Calculators > Databases lists every truss and corner block the app knows - 277 trusses from 17 makers (60 from the manufacturers' own current tables, the rest from the Truss Load Analyzer workbooks; 101 with native metric tables) and 51 corner blocks from Christie Lites and James Thomas Engineering - plus anything you have added.



The Databases page.

- **Search** by any part of a name, code or spec; filter to **Trusses** or **Corner blocks**, and by **maker**.
- Click a **Family** to see a truss together with the corner blocks that fit it (click the family chip to clear it).
- The **Source** column links to the maker's load table (trusses) or product page (corner blocks), or says *workbook* for data from Hall & Sogoian's workbook. Hover a row for where its numbers come from.
- Corner block weights marked *n/a* are not published - enter your own on the block in the rig.

Adding Your Own Truss

Open **Add your own truss**. Enter the manufacturer, model, weight per foot, maximum span, and the maximum UDL (total) and center point load (CPL) - these are used for every span up to the maximum. For values that change with span, paste one number per foot (1 ft to 100 ft; in metric, one per meter) into the optional UDL and CPL boxes, separated by commas. Tick **Data already includes the repetitive-use factor** if the maker's table already includes it. Click **Add truss**. It is saved in the units you entered it in and is marked *custom* in the pickers.

Adding Your Own Corner Block

Open **Add your own corner block**: maker, truss family, name, what it fits (e.g. `12x12`), number of ways, weight and size. Click **Add block**.

Custom Chain Hoists

At the bottom of the page: brand, model, capacity label (e.g. *1 Ton*), capacity, speed, body weight and chain weight per foot. Click **Add hoist**. A hoist with no speed uses the rig's default dynamic factor.

Custom entries can be deleted with the **×** on their row. They are kept in this browser and saved inside your rig files.

> **Custom data is your responsibility**
> Enter manufacturer figures exactly as published, and note whether the table already includes the repetitive-use factor. The app cannot check custom data.

18 How the App Calculates

You do not need this chapter to use the app, but it helps to know what the numbers mean and where they come from. The full technical description is in **File > About, method & version history** and on the calc sheet (section 2, *Basis of calculation*).

18.1 From Loads to Hoist Loads

1. Every truss is a beam carrying its point loads, its self weight (spread along it), its UDL and cable allowance, and anything passed in from trusses bolted to it or its corner blocks. Load factors (by category) and the *Add %* are applied as set in Rig settings.
2. The **whole-rig analysis** solves all the trusses together as beams meeting at the corner blocks - a direct-stiffness (finite element) analysis of the grid, often called a grillage. Trusses bend (through their chords), shear (through their diagonals) and twist; their stiffness is **estimated from the manufacturer's tables** unless the truss data gives real chord and diagonal sizes. Hoists are rigid supports at one level unless you give them a stiffness.
3. Because nobody publishes the stiffness of the joint through a corner block, the rig is solved with the joints **hinged** (the bolt passes vertical force only), **semi-rigid** (springs of 1, 4 and 16 × EI/L) and **rigid** (bending and twisting pass through). Each hoist's result is the **largest** of them, and each truss is checked with the joint model that loads it hardest (named in its *Checked with*).
4. A chain can only pull. If a hoist's reaction comes out negative, it is marked **Slack**, taken out, and the rig solved again. If that leaves a truss or assembly without enough support, it is **Unstable**.
5. Each hoist's **high hook static** load = its low hook load (the reaction) + Add % + the hoist body + chain (length × weight per foot) + hardware. The **high hook dynamic** load = static × the dynamic factor.

If the whole-rig analysis cannot run (a very large rig, or a load-path loop) the app falls back to the original workbook's **load-path method** - each truss solved on its own as a continuous beam (Clapeyron's three-moment equation), passing its reactions down the chain of bolted trusses to the hoists - and warns that it may under-estimate hoists in a grid.

18.2 Truss Checks

Check	Rule
Span	The span length is rounded **up** to the next whole foot (meter, for metric tables) and that row of the maker's table is read. Capacity = the table's center point load (CPL) × the repetitive-use factor k, reduced by the share of the uniform-load capacity already used by any UDL. The point loads on the span must not exceed it, and the span must not be longer than the table's maximum.
Cantilever	No longer than a quarter of the maximum span, and its load (point loads + UDL share + self weight, by default) no more than the CPL for a span four times the cantilever's length, × k.
Moment and shear	The bending moment and shear all along the truss (from the whole-rig analysis) against an allowable **estimated from the tables**: every table entry is a load the maker says the truss carries, so it shows a moment and a shear the truss can take. These are estimates - use the maker's published moment and shear where you have them.
Deflection	Each span's sag, measured from the straight line between its two supports, against the maker's published limit (past it: Overloaded) or the rig default L/160 (past it: a warning). Cantilever tips are shown, not checked.
Repetitive use	Capacities × 0.85 (ANSI) unless the data already includes it; the generic Universal trusses use 0.75.

18.3 Hoist Checks

A hoist's **status** compares its **static** high hook load with its rated capacity (as the original workbook does). Its dynamic load is flagged separately when it exceeds capacity. A dead hang is checked against its WLL.

18.4 Checked Against

The app's methods are checked in its test suite against the worked examples in Delbert L. Hall's *Rigging Math Made Simple*, and the whole-rig analysis against an independent 3D frame analysis of the example rigs in PyNite, the engine behind CalcForge.

19 Warnings and Messages

Warnings appear in step 4's Warnings tab, in the Trusses tab, in the banner, on the calc sheet and (for input mistakes) in Check rig. Most name the truss (and hoist) they are about - click the warning to select it. *Notes* are shown in gray and are not counted as warnings. In the messages below, *<Truss>* stands for the truss's name.

19.1 Truss Checks

Message (example)	Meaning	What to do
<Truss>: Span 2 - Overloaded (rigid joints)	The point loads on that span are more than the table allows. The joint model in brackets is the one that loads it hardest.	Shorten the span (add a hoist), move loads toward supports, or use a stronger truss.
... Span 1 - Too long	The span is longer than the table's maximum span.	Add a support.
... Left cantilever - Too long and overloaded	The overhang is longer than max span ÷ 4 and carries too much.	Move the end hoist out, or shorten/lighten the overhang.
<Truss>: Overloaded - bending moment 7,200 lb-ft is 106% of about 6,821 lb-ft allowed ...	The moment (or shear) is over the allowable estimated from the tables.	Usually a heavy load next to a support or a loaded neighboring span: move the load or add a hoist.
<Truss>: span 1 (24 ft) deflects about 2.1 in, more than span/160 ...	The span sags more than the rig default limit (a warning), or - starting "Overloaded -" - more than the maker's published limit.	Shorten or lighten the span.
<Truss>: its truss data is from the Truss Load Analyzer workbook, and the manufacturer's own table (...) allows less on its span 1 ...	The truss uses an older workbook row that allows more than the maker's current table at a span in your rig.	Switch the truss's Model to the maker's (MFG) row.
<Truss>: has no supports	No hoists and not bolted to anything.	Add hoists or bolt it.
<Truss>: unstable - only one support point is left, so the truss would tip ...	After slack hoists are taken out (or from the start) the truss has one support.	Add a hoist or bolted support, or move the load. Do not use its numbers.
Whole-rig analysis: ... (unstable)	An assembly of trusses is not held up (e.g. can tip about a line of hoists).	Add hoists out of line.

Message (example)	Meaning	What to do
Load-path loop or dependency on a loop: ...	Trusses are bolted to each other in a circle with nothing to take the load to a hoist first.	Give each truss in the loop a hoist, or break the loop.

19.2 Hoists

Message (example)	Meaning	What to do
<Truss> hoist: Overloaded (2,450 lb static, rigid joints)	High hook static load over the hoist's capacity (or a dead hang's WLL).	Share the load with another hoist, move loads, or use a bigger hoist.
... dynamic load exceeds capacity	Static is within capacity but static × dynamic factor is not.	Consider a bigger hoist or a slower one; follow the hoist maker's guidance.
... slack - the load would push this hoist up ...	The chain would go slack; the hoist carries only its own weight and the rest of the rig takes its share.	Move the hoist or the load (often a hoist beside a heavily loaded cantilever).
... goes slack with hinged joints only; the rigid joints result is used	Slack in some joint models only.	Be aware; the worst case is used.
10 hoists are level-sensitive (a 1/4" level error changes the load by more than 10% of the hoist's capacity ...)	Short, stiff spans: small level errors move a lot of load. The worst hoist and the others are listed.	Level carefully; consider load cells, a hoist stiffness, or an out-of-level tolerance.
Out of level by up to ±0.25 in ...: ... over capacity only because of the level allowance	With the tolerance set in Rig settings, these hoists would be overloaded if others ran off level.	Level more tightly, or add capacity.
... could unload completely and go slack on the low side	With the tolerance, these hoists could lose all their load.	Level carefully.
<Truss> hoist: hangs from X but is not under it on the plan ...	A hoist set to hang below a truss is not under that truss on the plan.	Move the hoist or the truss.
... no hoist model - its load is not checked against a capacity (note)	The hoist's model is "None".	Pick a model.

19.3 Bolting, Corner Blocks and Geometry

Message (example)	Meaning	What to do
North (JTE ...) is bolted to West CB1 (Christie ...): different manufacturers' truss parts don't bolt together ...	Bolted parts from different makers or truss families (the bolt is drawn red).	Use matching parts, or stack / clamp instead of bolting.
... generic, universal or custom parts - the app can't check that they bolt together (note)	The app cannot tell whether they match.	Check they are the same truss family.
Box truss: outside the manufacturer's corner-block rules - a box built with 90-degree corner blocks must be supported at every corner ...	The maker (e.g. Christie A Type) requires a hoist at every corner block of a box.	Hang a hoist at each corner block, or have the rig reviewed.
<Block>: ... joining 4 truss sections with no hoist at it. The maker's corner-block data allows only half of the truss table capacity ...	An unsupported block joining 4 or more sections (6 for a hub) - the checks shown still use full capacity.	Put a hoist at the block, or have it reviewed.
<Block>: 5 truss ends bolted to a 4-way block	More trusses on the block than it has ways.	Use a block with more ways, or change the layout.
<Block>: ... weight is not published - enter your own	The maker publishes no weight for the block.	Type a **Weight override** on the block.
<Truss> does not meet <Block>: it is 0.5 ft off, so the lengths around this loop do not close	In a box or grid, the lengths do not add up.	Check the pieces and corner blocks of the trusses in the loop.
X and Y are bolted to the same corner block on the same side, so they sit on top of each other	Two trusses on one face.	Bolt one to another block or flip it to the other side.
<Truss>: truss type not found in database / corner block type not found	The rig uses a custom entry this browser does not have.	Open a rig file that contains it, or add it again in Databases, then pick it.

19.4 Check Rig (Input Mistakes)

Message (example)	What to do
<Truss>: the load 'Spot' at 4 ft has no weight - enter it, or delete the load.	Type the weight, or delete the load.

Message (example)	What to do
... hoist at 10 ft and ... hoist at 10 ft are at the same point on the plan - they split its load between them. Is one of them a duplicate?	Delete the duplicate (or move one if there really are two).
Bolted assembly ...: only 2 hoists / all its hoists are in one line - a frame needs at least 3 hoists that are not in a line ...	Add hoists so the assembly cannot tip.
<Truss>: bolted to X at a point 6 in away from it on the plan ...	Check the positions of the bolted connection.
X and Y cross on the plan with no joint between them ... (note)	Bolt or stack them if they are joined; ignore if one runs above the other.

19.5 Other Messages

Message (example)	Meaning
Whole-rig analysis not available (...): hoist loads and truss checks are from the load-path method alone ...	The fallback method is in use; hoist loads in a grid may be under-estimated. Simplify the rig or remove the loop mentioned.
Hoist level offsets and the out-of-level tolerance need the whole-rig analysis ...	They are ignored while the fallback method is in use.
Measured loads: ... read 1,020 lb against 1,180 lb calculated (-14%) ...	An assembly's load cells differ from the calculation by more than 5%. Check the readings, the weights entered, and levels.
Whole-rig analysis does not balance ... - please report this rig	An internal self-check failed. Do not rely on the results; save the rig file and report it (see Credits).

20 Troubleshooting and Questions

My Rig Has Disappeared.

The app keeps the rig in this browser on this computer (see Where Your Work Is Kept). It is not there if you are in another browser, a private window, on another computer, or if the browser's data was cleared - or if you started a new rig or opened another file. Open your last saved `.rig.json` with **File > Open...**.

I Pressed New Rig by Mistake. Can I Undo It?

Yes - press `Ctrl` + `Z` (or click ↶). New rig, Open and Load example box are each one Undo step. This works only until you close or reload the page; after that, open your last saved file.

A Truss Says "Not Solved".

It has no support yet (add hoists or bolt it to a corner block), or it depends on something that is missing or on a loop of bolted trusses. The warning says which.

I Can't Drag a Truss.

Bolted trusses (and corner blocks) are placed by their block: move the truss that owns the block, or **Unbolt** it first.

"Bolt to..." Is Grayed Out.

There are no corner blocks on other trusses yet. Add a corner block to the truss you want to bolt *to* (its **Corner blocks** section).

The Truss Bolted to the Wrong Side or the Wrong End.

Use **Flip side** or **Bolt other end** in its Placement section, or **Unbolt** and bolt again with the options set (**its start / its end meets the block**, **extends north/south/east/west**).

A Load or Hoist Is at the Wrong Position.

Check its **From**: the same number means a different place from the start, the centerline or the end. On a centerline-measured truss, positions before the middle are negative.

Clicking + Hoist Added a Truss.

The add button adds to the *selected* truss; with nothing selected it adds a truss. Undo, select the truss, and click again.

The Hoist Loads Don't Add Up to What I Expect.

Open the hoist and look at **Where the load comes from**. In a grid, hoists share load through the trusses, so a hoist carries some weight from neighboring trusses (or has a little lifted off it). The joint model used is named underneath. **Compare analysis methods** in step 3 shows how much the joint assumption matters.

Why Are Hoist Loads So Different When I Move a Hoist a Little?

Short, stiff spans are level-sensitive. See Hoist Levels. Giving hoists a stiffness in Rig settings makes the analysis more realistic.

My Load Cells Don't Match the App.

Check you are comparing like with like (**Load cells read** in Rig settings: high hook or low hook), that every weight is entered (cable, clamps), and compare assembly totals rather than single hoists - levels and stiffness move load between hoists.

A Manufacturer Link Doesn't Open.

Links go to the maker's website and need an internet connection. Makers sometimes move their documents.

The App Looks Different from This Guide.

You are probably running a later version. This guide is for version 1.28.0. See **File > About** for what has changed.

Where Do I Report a Problem?

Save the rig as a file and send it, with a description, to the app's author (see Credits).

21 Keyboard and Mouse Reference

21.1 Anywhere (Not While Typing in a Box)

Key	Does
`1` `2` `3` `4`	Go to step 1 Structure, 2 Loads, 3 Hoists, 4 Results.
`Ctrl` + `Z`	Undo.
`Ctrl` + `Y` or `Ctrl` + `Shift` + `Z`	Redo.
`Delete` / `Backspace`	Delete the selected load, hoist or truss (a truss asks first).
`R`	Rotate the selected free truss 90° counterclockwise.
`F`	Fit the rig in the view.
`Esc`	Cancel "Bolt to..."; close Rig settings, a menu or the calc sheet.

21.2 In a Grid

Key	Does
Arrow keys	Move.
`Tab` / `Shift` + `Tab`	Next / previous editable cell (and finish an edit).
`Enter` / `Shift` + `Enter`	Down / up (and finish an edit).
Typing	Starts editing, replacing the value.
`F2`, double-click	Edit the value.
`Space`	Toggle a checkbox cell.
`Esc`	Cancel the edit; again to leave the grid; clears a multi-selection.
`Ctrl` + `D`	Fill down from the row above (or from the top selected hoist).
`Ctrl` + `C` / `Ctrl` + `V`	Copy the cell / paste a cell or rows from a spreadsheet.
`Delete`, `Shift` + `Backspace`	Delete the row (or all selected hoists).
`Shift` +click, `Ctrl` +click, `Shift` + `↑` `↓`, `Ctrl` + `A`	Hoists grid: select several hoists.

21.3 In a Search List

`↑` `↓` choose, `Enter` pick, `Tab` pick and move on, `Esc` close.

21.4 Mouse on the Plan

Action	Does
Click	Select a truss, corner block or hoist; empty space selects the rig.
Drag a free truss	Move it (ends snap; `Alt` = no snap).
Drag a hoist	Slide it along its truss in 1" steps.
Drag empty space	Pan.
Wheel	Zoom around the pointer.
Hover	Tooltip with details.

21.5 Mouse in 3D

Drag = orbit, `Shift`+drag or right-drag = pan, wheel = zoom, click = select.

21.6 Help and Search in This Guide

The app's **Help** menu opens the Quick Start and this guide (HTML or PDF) in a new browser tab. It looks for them in the `docs/manual` folder next to the app; if you have only the app file, use **Help > All manuals online**.

In the HTML copy of this guide, `/` or `Ctrl`+`K` jumps to the search box; `Enter` opens the first match; `Esc` clears the search.

22 Glossary

Term	Meaning
Add %	An optional extra percentage on the load at every hoist (Rig settings).
Allowable (moment, shear)	The bending moment or shear a truss may carry, estimated in this app from the maker's load tables.
Assembly	A group of trusses joined together by bolts, stacking or corner blocks.
Bolted	Joined to another truss through a corner block (or bolted end to a truss). A bolted truss is positioned by its block.
Calc sheet	The printable calculation, for independent review of the software's work.
Cantilever	The part of a truss past its outermost support. Max length = max span ÷ 4.
Carrier	A truss that other hoists hang below.
Centerline (CL)	The middle of a truss line; positions measured from it are negative toward the start.
Check rig	The list of likely input mistakes, in the rig panel.
Clamp	The weight of the fixture's clamp, added to the load.
Corner block	A block trusses bolt to at right angles; part of one truss line in the app.
CPL	Center point load: the maker's allowable single load in the middle of a span.
Dead hang	A support of wire rope and hardware with no hoist.
Deflection	How far a span sags, measured from the line between its supports; limit given as span/ratio (e.g. L/160).
Derate / repetitive-use factor	The multiplier on table capacities for repeated use (0.85 per ANSI unless included).
Dynamic factor (DLF)	Multiplier for loads while a hoist moves: speed (fpm) ÷ 60 + 1.
Fingerprint	A short code on the calc sheet that changes whenever any input changes.
Grillage	The engineering name for the whole-rig analysis of a grid of beams.
Hangs from	What holds a truss up (its hoists, blocks and trusses), or where a hoist hangs from (the structure, or below a truss).
High hook load	Low hook load + Add % + hoist, chain and hardware: what the structure above carries. Static and dynamic.

Term	Meaning
Hinged / semi-rigid / rigid joints	The three ways the whole-rig analysis models the joints through corner blocks; the worst is used.
Hot spots	A color mode showing where along each truss it works hardest.
Level offset	A designed height difference for one hoist (+ higher).
Level sensitivity	The change in a hoist's load if it runs 1/4" high or low.
Line	A truss line: truss pieces plus corner blocks, end to end.
Load factor	A multiplier on loads of one category (Rig settings).
Load-path method	The original workbook's method, used as a fallback when the whole-rig analysis cannot run.
Low hook load	What the truss hangs on a hoist's hook (the reaction, R).
MFG	Truss data from the manufacturer's own current load table.
Mirror	A load's twin at the mirror position about the centerline (loads); a mirrored copy (hoists).
Pieces	The truss sections in a line, not counting corner blocks.
Slack	A hoist whose chain would go slack because the load pushes it up.
Span	The part of a truss between two supports.
Stack / clamp	A truss resting on top of (or clamped under) another where they cross.
Truss	In the app: one straight truss line.
UDL	Uniformly distributed load: a weight spread evenly along a truss (typed as a total).
Unstable	Not held up by enough supports; would tip.
Whole-rig analysis	The stiffness analysis of the whole rig that gives the hoist loads and the forces the checks use.
Workbook	Hall & Sogoian's *Truss Load Analyzer - EOT* Excel workbook, the original program.
Workload	Load as a percentage of what is allowed.
WLL	Working load limit.

23 Credits, License and Disclaimer

23.1 Credits

Truss Grid Analyzer, copyright © 2026 **G.E. Simmons Falk**.

An expansion of **Truss Load Analyzer - EOT** by **Delbert L. Hall and Jon Sogoian**, who are the originators of the program, for indeterminate grids. Developed with the knowledge and permission of Delbert L. Hall. Its method and its truss, chain hoist and fixture data come from their workbook. Manufacturer data is from Christie Lites, James Thomas Engineering, Tomcat, Tyler Truss and the other makers named in the Databases page. Thanks to **Issy Stadler** for testing help.

23.2 License

Truss Grid Analyzer is free. Anyone may use, copy, modify and distribute it free of charge, provided they keep the license file with every copy, credit Delbert L. Hall and Jon Sogoian as the originators of the program and G.E. Simmons Falk as the author of Truss Grid Analyzer, update the version history to show who changed what and when, and **never charge** for it. The full terms are in the `LICENSE.md` file that comes with the app.

23.3 Disclaimer

This tool is intended for use by entertainment rigging professionals to assist them in making rigging decisions. Every attempt has been made to be accurate. However, the authors of the original program and of this expansion are not responsible for errors in the program or the results of its use. **Users are responsible for verifying the results before using them to make rigging decisions.**

23.4 About This Guide

User Guide for Truss Grid Analyzer version 1.28.0, 12 October 2026. The guide is revised on request, not with every release of the app.